



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

Department of Computer Science
Faculty of Computing

GROUP PROJECT

(Benchmarking of Computer Performance)

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Section: 03

Video Link : <https://youtu.be/IeZogO0Wa24>

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1.0 Introduction

1.1 Problem background

In today's fast-paced computing world, it is essential to evaluate the performance of hardware components, especially central processing units (CPUs). As technology advances and high-performance applications become more common, users need reliable methods to measure and compare different CPUs. This is where benchmarking comes in. CPU benchmarking involves running standardized tests to measure a CPU's performance under controlled conditions. These tests provide metrics that allow for an objective comparison of different systems, revealing their strengths and weaknesses.

Benchmarking is important for hardware manufacturers, consumers, and IT professionals. Manufacturers use benchmarks to improve their products and highlight their advantages. Consumers rely on benchmarking data to make informed purchasing decisions, ensuring they choose systems that meet their needs. IT professionals use benchmarking to optimize system performance, diagnose issues, and manage resources effectively.

Many benchmarking tools are available, each with unique features. These tools measure various aspects of CPU performance, such as processing speed, multi-threading efficiency, and thermal behavior. By using these tools, users can understand how their systems perform under different workloads.

Benchmarking also provides valuable hands-on learning experiences. Students can apply theoretical knowledge to practical scenarios, exploring how different hardware configurations and software environments impact CPU performance. This project uses four free benchmarking tools—CPU-Z, Geekbench, and Windows Task Manager, MiniTool Partition Wizard—to evaluate the performance of four different laptops. Through this process, students will learn to collect, analyze, and interpret benchmarking data, gaining valuable skills in hardware performance evaluation.

The project consists of two parts: first, using the benchmarking tools to measure laptop performance, and second, running a standardized assembly program on each laptop to further evaluate their efficiency in handling specific tasks. This approach ensures a comprehensive understanding of both general and specific aspects of CPU performance.

By the end of this project, students will better understand CPU performance evaluation and the practical applications of benchmarking in both consumer and professional contexts.

1.2 Aim

The primary aim of this project is to compare the benchmarking results of four different laptops using four distinct benchmarking tools. Additionally, the project seeks to measure the efficiency of an assembly program run on each of these laptops, highlighting the differences in CPU performance.

1.3 Objectives

- **Utilize Four Benchmarking Tools:** Employ CPU-Z, MiniTool Partition Wizard, Geekbench, and Windows Task Manager to benchmark the performance of four different laptops.
- **Analyze Benchmarking Results:** Collect and compare the data from these tools to identify each CPU's strengths and weaknesses across various computing tasks.
- **Execute Assembly Program:** Run a standardized assembly program on all four laptops to evaluate the efficiency of their CPUs in terms of running time and elapsed time.
- **Compare Performance Metrics:** Assess the collected performance metrics to draw comprehensive conclusions about the capabilities of the CPUs in different scenarios.

1.4 Scopes (computer specification & benchmarking tools features)

For computers to be benchmarked, we use ASUS Vivobook 15, Dell Inc. Inspiron 16 5630, Huawei Matebook D15 2022 and Lenovo IdeaPad Slim 5 16IAH8.

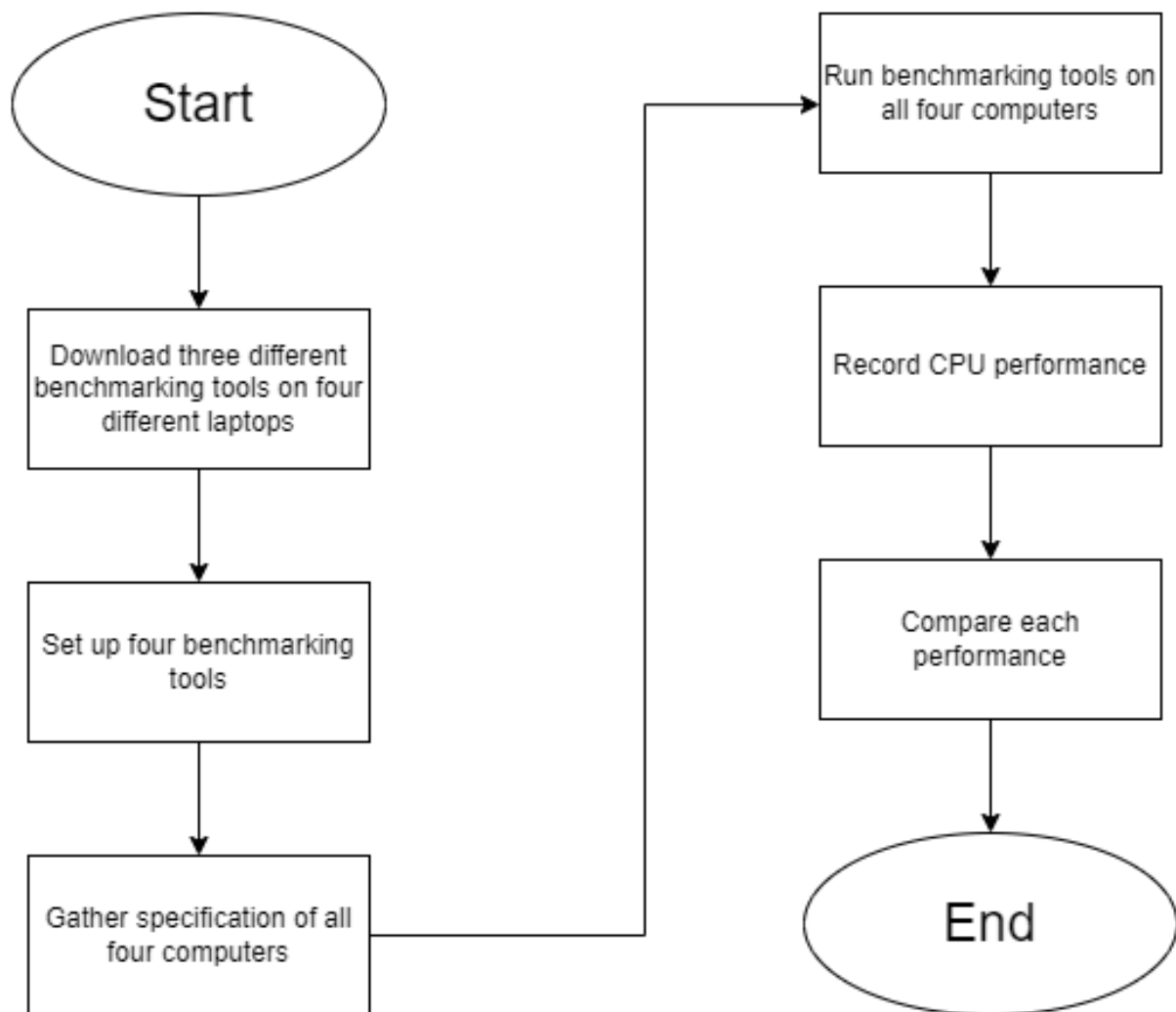
Specifications	Computer Type			
	ASUS Vivobook 15	Dell Inc. Inspiron 16 5630	Huawei Matebook D15 2022	Lenovo IdeaPad Slim 5 16IAH8
Processor Type	Intel Core i5-1135G7	Intel Core i7-360P	Intel Core i5-10210U	Intel Core i5-12450H
RAM Size	8 GBytes	16 GBytes	8 GBytes	16 GBytes
GPU Type	Intel® Iris Xe Graphics	Intel® Iris Xe Graphics	Intel® UHD Graphics	Intel® UHD Graphics
Motherboard	ASUSTEK COMPUTER Inc.X513EP	Dell Inc. 065RWX	HUAWEI BOHB-WAX9-P CB-B2	LENOVO LNVNB 161216
Bus Specification	PCI-Express 3.0 (8.0 GT/s)	PCI-Express 4.0 (16.0 GT/s)	PCI-Express 3.0 (8.0 GT/s)	PCI-Express 4.0 (16.0 GT/s)

2.0 Project - Part 1

2.1 Overview

To investigate the performance of three CPUs and make an analysis of the performance data and make a comparison to the three CPUs.

2.2 Flowchart of the project



In this project, we are using the CPU-Z, Geekbench, Windows Task Manager and the MiniTool Partition Wizard Free 12.8 to benchmark the computers.

1. **CPU-Z** is a freeware system profiling and monitoring application for Microsoft Windows and Android that detects the central processing unit (CPU), random-access memory (RAM), motherboard chipset, GPU and other hardware features of a modern personal computer or Android device. It gathers and displays detailed information about these key system components and the hardware in a device.

Here are some of the features of the CPU-Z benchmarking tool:

- Displays detailed information about the CPU, including code name, package, and technology
- Shows core voltage, clock speed, and multiplier
- Details about the CPU caches, including L1, L2, and L3 cache sizes and speeds
- Provides information about the motherboard, including manufacturer, model, chipset
- Displays details about the system memory, including type, size, and timings
- Displays information about the GPU

2. **Geekbench** is a cross-platform benchmarking tool that measures the performance of your computer or mobile device. It provides a comprehensive set of benchmarks that give insights into the performance of your CPU, GPU, and memory. It is available for a variety of platforms, including Windows, macOS, Linux, iOS, and Android.

Here are some of the features of the Geekbench benchmarking tool:

- Allows comparison of performance across different devices and operating systems
- Measures the performance of a single core in your processor
- Evaluates the performance when multiple cores are utilized
- Measures the battery performance
- Tests the compute performance of GPUs and other accelerators
- Provides a database of results from various devices for comparison
- Supports 64-bit benchmarks, providing more accurate and relevant performance metrics

3. **Windows Task Manager** is a system monitoring tool included in Microsoft Windows operating systems. It provides detailed information about the computer's performance,

running applications, processes, and system resources. It is a powerful tool for managing and monitoring system performance, diagnosing issues, and controlling applications and services running on the computer.

Here are some of the primary features of Windows Task Manager:

- Displays all running applications and background processes with resource usage details
- Shows real-time graphs and statistics for CPU, memory, disk, network, and GPU usage
- Tracks resource usage over time for Windows Store apps
- Lists and manages programs that run at system startup
- Displays active user sessions and their resource usage
- Provides in-depth information about each running process
- Lists system services and allows starting, stopping, or restarting them

4. **MiniTool Partition Wizard Free 12.8** is a comprehensive disk partition management tool designed to provide users with an intuitive interface and powerful features to manage their hard drive partitions effectively. It supports a wide range of partitioning tasks and disk-related operations, making it a versatile utility for both novice and advanced users.

Here are some of the features of the MiniTool Partition Wizard Free 12.8 benchmarking tool:

- Create, delete, format, move, resize, and merge partitions
- Convert disks between MBR and GPT formats without data loss
- Convert file systems between NTFS and FAT32 without reformatting
- Recover lost or deleted partitions on both MBR and GPT disks
- Clone entire disks or individual partitions for data backup and migration
- Measure and analyze disk performance to identify potential issues
- Check the health of a disk by scanning for bad sectors
- Migrate the operating system to a new SSD or HDD without reinstalling
- Align partitions to improve performance, especially for SSDs
- Manage dynamic disks and volumes, including creation, deletion, and conversion to basic disks

2.3 Benchmarking Result

CPU-Z

Benchmark	Computer type			
	ASUS Vivobook 15	Dell Inc. Inspiron 16 5630	Huawei Matebook D15 2022	Lenovo 83BG
CPU Technology	10 nm	10 nm	14 nm	10 nm
Number of Core	4	12	4	8
Number of threads	8	16	8	12
Core speed	3791.43 MHz	1895.36 MHz	3492.30 MHz	1695.85 MHz
Bus Speed	99.77 MHz	99.76 MHz	99.78 MHz	99.76 MHz
RAM Size	8 GBytes	16 GBytes	8 GBytes	16 GBytes
GPU Size	2 GBytes	-	-	-

CPU-Z

CPU

Mainboard

Memory

SPD

Graphics

Bench

About

Processor

Name

Intel Core i5 1135G7

Code Name

Tiger Lake-U

Max TDP

28.0 W

Package

Socket 1449 FCBGA

Technology

10 nm

Core VID

1.092 V

Specification

11th Gen Intel® Core™ i5-1135G7 @ 2.40GHz

Family

6

Model

C

Stepping

1

Ext. Family

6

Ext. Model

8C

Revision

B1

Instructions

MMX, SSE, SSE2, SSE3, SSSE3, SSE4.1, SSE4.2, EM64T, VT-x, AES, AVX, AVX2, AVX512, FMA3, SHA

Clocks (Core #0)

Core Speed

3791.43 MHz

Multiplier

x 38.0 (4.0 - 42.0)

Bus Speed

99.77 MHz

Rated FSB

Cache

L1 Data

4 x 48 KBytes

12-way

L1 Inst.

4 x 32 KBytes

8-way

Level 2

4 x 1.25 MBytes

20-way

Level 3

8 MBytes

8-way

Selection

Socket #1

Cores

4

Threads

8

CPU-Z

Ver. 2.09.0.x64

Tools

▼

Validate

Close



CPU-Z

CPU

Mainboard

Memory

SPD

Graphics

Bench

About

Processor

Name

Intel Core i5 10210U

Code Name

Comet Lake-U/Y

Max TDP

15.0 W

Package

Socket 1528 FCBGA

Technology

14 nm

Core VID

0.980 V

Specification

Intel® Core™ i5-10210U CPU @ 1.60GHz

Family

6

Model

E

Stepping

C

Ext. Family

6

Ext. Model

8E

Revision

C0

Instructions

MMX, SSE, SSE2, SSE3, SSSE3, SSE4.1, SSE4.2, EM64T, VT-x, AES, AVX, AVX2, FMA3



Clocks (Core #0)

Core Speed

3492.30 MHz

Multiplier

x 35.0 (4.0 - 42.0)

Bus Speed

99.78 MHz

Rated FSB

Cache

L1 Data

4 x 32 KBytes

8-way

L1 Inst.

4 x 32 KBytes

8-way

Level 2

4 x 256 KBytes

4-way

Level 3

6 MBytes

12-way

Selection

Socket #1

Cores

4

Threads

8

CPU-Z

Ver. 2.09.0.x64

Tools

Validate

Close

ASUS VIVOBOK 15

CPU-Z

CPU

Mainboard

Memory

SPD

Graphics

Bench

About

Processor

Name

Intel Core i5 10210U

Code Name

Comet Lake-U/Y

Max TDP

15.0 W

Package

Socket 1528 FCBGA

Technology

14 nm

Core VID

0.980 V



Specification

Intel® Core™ i5-10210U CPU @ 1.60GHz

Family

6

Model

E

Stepping

C

Ext. Family

6

Ext. Model

8E

Revision

C0

Instructions

MMX, SSE, SSE2, SSE3, SSSE3, SSE4.1, SSE4.2, EM64T, VT-x, AES, AVX, AVX2, FMA3

Clocks (Core #0)

Core Speed

3492.30 MHz

Multiplier

x 35.0 (4.0 - 42.0)

Bus Speed

99.78 MHz

Rated FSB

Cache

L1 Data

4 x 32 KBytes

8-way

L1 Inst.

4 x 32 KBytes

8-way

Level 2

4 x 256 KBytes

4-way

Level 3

6 MBytes

12-way

Selection

Socket #1

Cores

4

Threads

8

CPU-Z

Ver. 2.09.0.x64

Tools

Validate

Close

Dell Inc.Inspiron 16 5630

CPU-Z

CPU

Mainboard

Memory

SPD

Graphics

Bench

About

Processor

Name

Intel Core i5 12450H

Code Name

Alder Lake

Max TDP

45.0 W

Package

Socket 1744 FCBGA

Technology

10 nm

Core VID

0.780 V

intel

CORE

i5

Specification

12th Gen Intel® Core™ i5-12450H

Family

6

Model

A

Stepping

3

Ext. Family

6

Ext. Model

9A

Revision

L0

Instructions

MMX, SSE, SSE2, SSE3, SSSE3, SSE4.1, SSE4.2, EM64T, AES, AVX, AVX2, AVX-VNNI, FMA3, SHA

Clocks (P-Core #0)

Core Speed

1695.85 MHz

Multiplier

x 17.0 (4.0 - 44.0)

Bus Speed

99.76 MHz

Rated FSB

Cache

L1 Data

4 x 48 KB + 4 x 32 KB

L1 Inst.

4 x 32 KB + 4 x 64 KB

Level 2

4 x 1.25 MB + 2 MBytes

Level 3

12 MBytes

Selection

Socket #1

Cores

4P + 4E

Threads

12

CPU-Z

Ver. 2.09.0.x64

Tools

Validate

Close

Huawei Matebook D15 2022

Lenovo 83BG

GEEKBENCH

Benchmark	Computer type			
	ASUS Vivobook 15	Dell Inc. Inspiron 16 5630	Huawei Matebook D15 2022	Lenovo 83BG
CPU				
Name	Intel Core i5-1135G7	Intel Core i7-1360P	Intel Core i5-10210U	Intel Core i5-12450H
Topology	1 Processor, 12 Cores, 16 Threads	1 Processor, 4 Cores, 8 Threads	1 Processor, 4 Cores, 8 Threads	1 Processor, 8 Cores, 12 Threads
Base Frequency	2.40 GHz	2.20 GHZ	2.09 GHz	2.00 GHz
Maximum frequency	4193 MHz	4987 MHz	4189 MHz	4389 MHz
Single-core Score	1739	2336	1207	1078
Multiple-core Score	4250	9596	3335	4653
Codename	Tiger Lake-U	Raptor Lake	Comet Lake - U/Y	Alder Lake
L1 Instruction Cache	32.0 KB x 4	32.0 KB x 8	32.0 KB x 4	32.0 KB x 6
L1 Data Cache	48.0 KB x 4	48.0 KB x 8	32.0 KB x 4	48.0 KB x 6
L2 Cache	1.25 MB x 4	1.25 MB x 2	256 KB x 4	1.25 MB x 1
L3 Cache	8.00 MB x 1	18.0 MB x 1	6.0 MB x 1	12.0 MB x 1
Memory				
Size	11.69 GB	15.69 GB	7.84 GB	16 GB
Transfer Rate	2660 MT/s	4788 MT/s	2392 MT/s	2394 MT/s
Type	DDR4 SDRAM	DDR5 SDRAM	DDR4 SDRAM	DDR5 SDRAM
Channels	2	4	2	4

ASUSTeK COMPUTER INC. VivoBook_ASUSLaptop X513EP_A513EP

1739	4250
Single-Core Score	Multi-Core Score

Desktop 8.0 for Windows ARM2

Result Information	
Upload Date	June 29 2024 01:43 PM
Views	2

System Information

System Information	
Operating System	Microsoft Windows 11 Home Single Language (24H2)
Model	ASUSTeK COMPUTER INC. VivoBook_ASUSLaptop X513EP_A513EP
Motherboard	ASUSTeK COMPUTER INC. X513EP
Power Plan	ASUS Recommended

CPU Information	
Name	Intel Core i5-1350P
Topology	1 Processor, 4 Cores, 8 Threads
Identifier	Genualderm Family 8 Model 140 Stepping 1
Base Frequency	2.40 GHz
Cluster 1	4 Cores
Maximum Frequency	4.50 GHz
Package	Soclet 1345 FCBGA
Codename	Tiger Lake-U
L1 Instruction Cache	32.0 KiB x 4
L1 Data Cache	48.0 KiB x 4
L2 Cache	1.25 MiB x 4
L3 Cache	8.00 MiB x 1

Dell Inc. Inspirin 16 5630

2336	9596
Single-Core Score	Multi-Core Score

Desktop 8.0 for Windows ARM2

Upload Date	June 29, 2024 10:05 AM
Views	7

System Information

Operating System	Microsoft Windows 11 Home Single Language (24H2)
Model	Dell Inc. Inspiron 16 5630
Motherboard	Dell Inc. 0000WXX
Power Plan	Balanced

CPU Information	
Name	Intel Core i7-1350P
Topology	1 Processor, 12 Cores, 16 Threads
Identifier	Genualderm Family 8 Model 180 Stepping 2
Base Frequency	2.20 GHz
Cluster 1	4 Cores
Cluster 2	4 Cores
Maximum Frequency	4.60 GHz
Package	Soclet 1345 FCBGA
Codename	Raptor Lake
L1 Instruction Cache	32.0 KiB x 4
L1 Data Cache	48.0 KiB x 4
L2 Cache	1.25 MiB x 2
L3 Cache	18.0 MiB x 1

HUAWEI BOHB-WX9

1207	3335
Single-Core Score	Multi-Core Score

Desktop 8.0 for Windows ARM2

Upload Date	June 29, 2024 09:22 AM
Views	9

System Information

System Information	
Operating System	Microsoft Windows 11 Home Single Language (24H2)
Model	HUAWEI BOHB-WX9
Motherboard	HUAWEI BOHB-WX9/FCB 82
Power Plan	Balanced

CPU Information	
Name	Intel Core i5-1215U
Topology	1 Processor, 4 Cores, 8 Threads
Identifier	Genualderm Family 8 Model 142 Stepping 12
Base Frequency	2.00 GHz
Cluster 1	4 Cores
Maximum Frequency	4.50 GHz
Package	Soclet 1305 FCBGA
Codename	Comet Lake-U7
L1 Instruction Cache	32.0 KiB x 4
L1 Data Cache	32.0 KiB x 4
L2 Cache	200 KiB x 4
L3 Cache	8.00 MiB x 1

LENOVO 83BG

1078	4653
Single-Core Score	Multi-Core Score

Desktop 8.0 for Windows ARM2

User	cheryl
Upload Date	June 29 2024 03:34 PM
Views	12

System Information

System Information	
Operating System	Microsoft Windows 11 Home Single Language (24H2)
Model	LENOVO 83BG
Motherboard	LENOVO L880R01210
Power Plan	Balanced

CPU Information	
Name	Intel Core i5-1250P
Topology	1 Processor, 8 Cores, 12 Threads
Identifier	Genualderm Family 8 Model 134 Stepping 2
Base Frequency	2.00 GHz
Cluster 1	4 Cores
Cluster 2	4 Cores
Maximum Frequency	4.00 GHz
Package	Soclet 1345 FCBGA
Codename	Alder Lake
L1 Instruction Cache	32.0 KiB x 8
L1 Data Cache	48.0 KiB x 8
L2 Cache	1.25 MiB x 1
L3 Cache	12.0 MiB x 1

Memory Information	
Size	16.00 GB
Transfer Rate	2880 MT/s
Type	DDR4 SDRAM

Single-Core Performance

Single-Core Score	1739
File Compression	1804
Navigation	1888
HTML5 Browser	939 navigation
PDF Renderer	1736
Photo Library	1718
Clang	1983
Text Processing	1812
Asset Compression	1885
Object Detection	1876
Background Blur	2001
Horizon Detection	2040
Object Removal	1888
HDR	1918
Photo Filter	2046
Ray Tracer	1918
Structure from Motion	1904

Memory Information	
Size	16.00 GB
Transfer Rate	4700 MT/s
Type	DDR5 SDRAM
Channels	4

Single-Core Performance

Single-Core Score	2336
File Compression	2217
Navigation	2171
HTML5 Browser	2089
PDF Renderer	2240
Photo Library	2247
Clang	2324
Text Processing	2224
Asset Compression	2219
Object Detection	2275
Background Blur	2323
Horizon Detection	2340
Object Removal	1925
HDR	2243
Photo Filter	2348
Ray Tracer	2200
Structure from Motion	2207

Memory Information	
Size	7.04 GB
Transfer Rate	2380 MT/s
Type	DDR5 SDRAM

Single-Core Performance

Single-Core Score	1207
File Compression	1429
Navigation	1527
HTML5 Browser	1037
PDF Renderer	1246
Photo Library	1280
Clang	1324
Text Processing	1246
Asset Compression	1288
Object Detection	800
Background Blur	1407
Horizon Detection	1544
Object Removal	1280
HDR	1281
Photo Filter	1427
Ray Tracer	1198
Structure from Motion	1478

Memory Information	
Size	16.00 GB
Transfer Rate	2364 MT/s
Type	DDR5 SDRAM

Single-Core Performance

Single-Core Score	1078
File Compression	1481
Navigation	1587
HTML5 Browser	937 navigation
PDF Renderer	1022
Photo Library	1050
Clang	1500
Text Processing	1077
Asset Compression	1090
Object Detection	800
Background Blur	1497
Horizon Detection	1614
Object Removal	800
HDR	1100
Photo Filter	1280
Ray Tracer	1074
Structure from Motion	1121

Multi-Core Performance

Multi-Core Score	4250
File Compression	3928
Navigation	3410
HTML5 Browser	2761
PDF Renderer	3540
Photo Library	4017
Clang	4460
Text Processing	3221
Asset Compression	3767
Object Detection	3239
Background Blur	3789
Horizon Detection	3627
Object Removal	3758
HDR	3884
Photo Filter	3530
Ray Tracer	3865
Structure from Motion	4006

Multi-Core Performance

Multi-Core Score	9596
File Compression	6999
Navigation	12040
HTML5 Browser	12112
PDF Renderer	12442
Photo Library	11417
Clang	11841
Text Processing	2412
Asset Compression	12642
Object Detection	6789
Background Blur	9072
Horizon Detection	10149
Object Removal	6989
HDR	10042
Photo Filter	9617
Ray Tracer	16233
Structure from Motion	11532

Multi-Core Performance

Multi-Core Score	3335
File Compression	1742
Navigation	2684
HTML5 Browser	2490
PDF Renderer	2374
Photo Library	2712
Clang	2784
Text Processing	1620
Asset Compression	1650
Object Detection	1476
Background Blur	1605
Horizon Detection	2347
Object Removal	2684
HDR	4194
Photo Filter	2840
Ray Tracer	5839
Structure from Motion	4227

Multi-Core Performance

Multi-Core Score	4653
File Compression	3839
Navigation	3491
HTML5 Browser	4924
PDF Renderer	3691
Photo Library	3623
Clang	5280
Text Processing	1300
Asset Compression	3676
Object Detection	3238
Background Blur	3117
Horizon Detection	3676
Object Removal	4228
HDR	4746
Photo Filter	3001
Ray Tracer	6146
Structure from Motion	4042

Task Manager

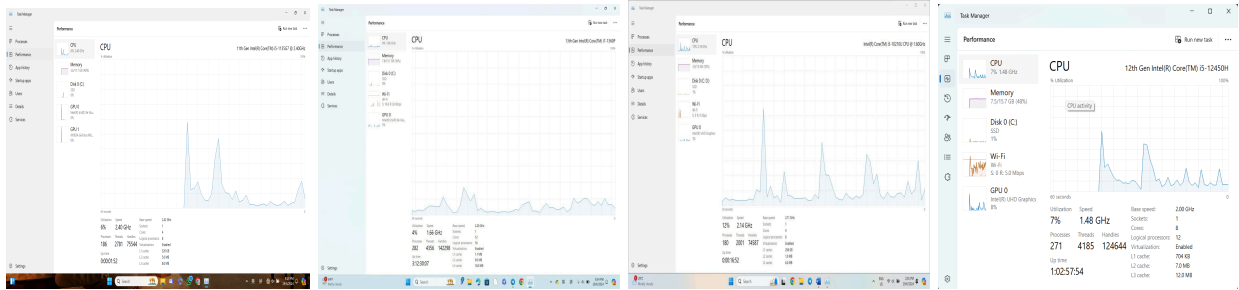
Performance	Computer type			
	ASUS Vivobook 15	Dell Inc. Inspiron 16 5630	Huawei Matebook D15 2022	Lenovo IdeaPad Slim 5 16IAH8
CPU Base Speed	2.42 GHz	2.20 GHz	2.11 GHz	2.00 GHz
CPU Sockets	1	1	1	1
Cpu Cores	4	12	4	8
CPU Logical Processors	8	16	8	12
CPU Virtualization	Enabled	Enabled	Enabled	Enabled
CPU L1 Cache	320 KB	1.1 MB	256 KB	704 KB
CPU L2 Cache	5.0 MB	9.0 MB	1.0 MB	7.0 MB
CPU L3 Cache	8.0 MB	18.0 MB	6.0 MB	12.0 MB
Memory Speed	2667 MHz	4800 MHz	2400 MHz	4800 MHz
Memory Slots Used	2 of 8	8 of 8	2 of 2	2 of 2
Memory Form Factor	SODIMM	Row of chips	SODIMM	Row of chips
Memory Hardware Reserved	318 MB	315 MB	167 MB	276 MB

Load	Low Load				Middle Load			
Computer type	ASUS Vivobook 15	Dell Inc. Inspiron 16 5630	Huawei Matebook D15 2022	Lenovo IdeaPad Slim 5 16IAH8	ASUS Vivobook 15	Dell Inc. Inspiron 16 5630	Huawei Matebook D15 2022	Lenovo IdeaPad Slim 5 16IAH8
CPU Usage	7%	4%	40%	7%	3%	5%	5%	12%
CPU Utilization	6%	4%	12%	7%	3%	17%	5%	12%
CPU Speed	2.40 GHz	1.66 GHz	2.14 GHz	1.48 GHz	2.30 GHz	1.48 GHz	1.24 GHz	1.98 GHz
CPU Processes	186	282	180	271	231	285	179	294
CPU Threads	2781	4356	2001	4185	2997	4459	2037	4437
CPU Handles	75544	142298	74587	124644	121951	147673	75020	146888
CPU Up time	0:00:01:52	3:12:30:07	0:00:16:52	1:02:57:54	16:01:44:52	3:12:27:57	0:00:17:46	1:03:02:56
Memory Usage	54%	52%	50%	48%	66%	61%	65%	61%
Memory In use (compressed)	6.3 GB (379 MB)	8.0 GB (655MB)	3.8 GB (42.7MB)	7.2 GB (136MB)	7.7 GB (415 MB)	9.5 GB (745MB)	5.1 GB (45.8MB)	9.5 GB (379 MB)
Memory Available	5.4 GB	7.6 GB	4.0 GB	8.2 GB	3.9 GB	6.2 GB	2.7 GB	6.1 GB
Memory Committed	8.0/17.7 GB	11.5/26.2 GB	3.9/15.3 GB	7.9/18.1 GB	9.8/18.9 GB	13.6/26.2 GB	5.2/15.3 GB	10.7/18.1 GB
Memory Cached	3.0 GB	4.0 GB	3.1 GB	5.6 GB	2.7 GB	3.5 GB	2.6 GB	5.4 GB

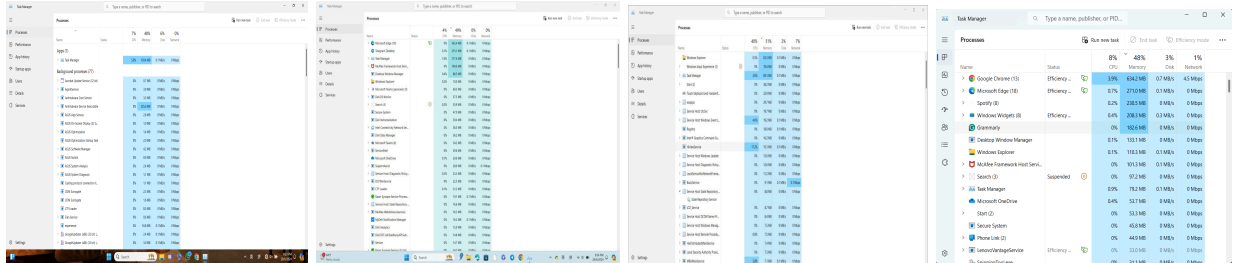
Memory Paged pool	432 MB	659 MB	485 MB	435 MB	719 MB	664 MB	516 MB	487 MB
Memory Non-paged pool	664 MB	1.1 GB	373 MB	512 MB	896 MB	1.1 GB	377 MB	525 MB

Load	High Load			
Computer type	ASUS Vivobook 15	Dell Inc. Inspiron 16 5630	Huawei Matebook D15 2022	Lenovo IdeaPad Slim 5 16IAH8
CPU Usage	13%	14%	14%	35%
CPU Utilization	2%	5%	5%	27%
CPU Speed	1.78 GHz	1.53 GHz	1.18 GHz	2.58 GHz
CPU Processes	255	347	224	323
CPU Threads	3585	3585	2947	5117
CPU Handles	139190	183334	99316	166617
CPU Up time	16:01:43:30	3:12:22:01	0:00:19:04	1:03:10:53
Memory Usage	77%	80%	77%	79%
Memory In use (compressed)	9.0 GB (328 MB)	12.6 GB (870 MB)	6.0 GB (91.0MB)	12.5 GB (351MB)
Memory Available	2.7 GB	3.1GB	1.8 GB	3.2 GB
Memory Committed	11.5/18.9 GB	17.1/26.2 GB	6.8/15.3 GB	14.5/18.1 GB
Memory Cached	2.7 GB	3.1 GB	1.6 GB	3.2 GB
Memory Paged pool	679 MB	670 MB	506 MB	485 MB
Memory Non-paged pool	872 MB	1.1 GB	380 MB	561 MB

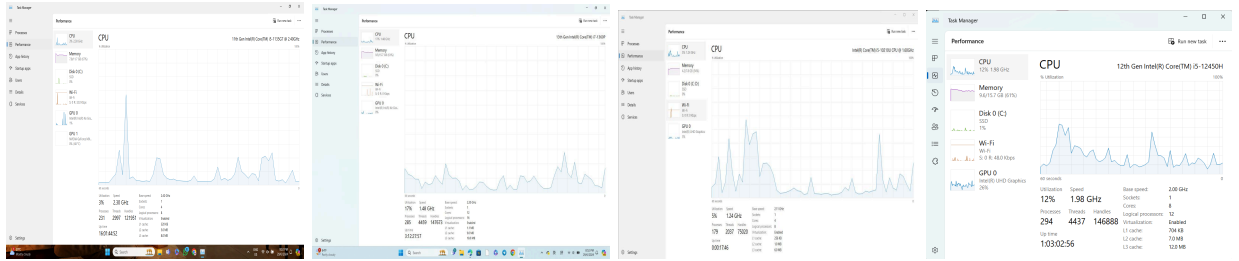
Low Load Performance



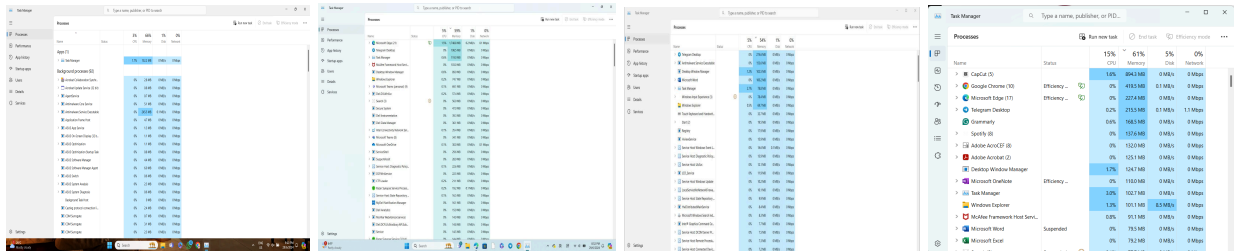
Low Load Process



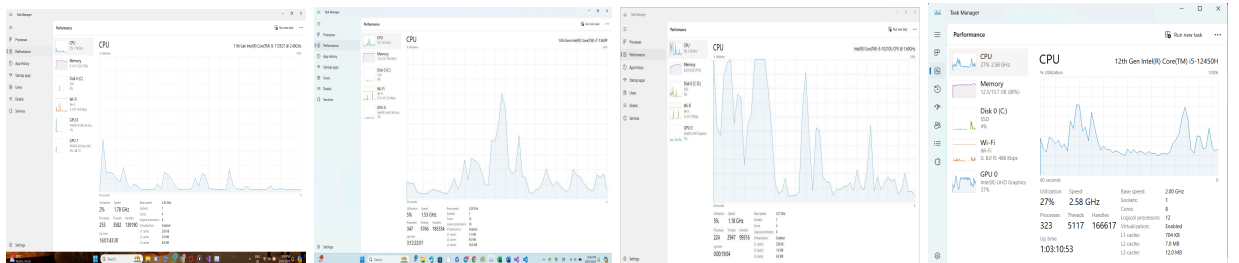
Middle Load Performance



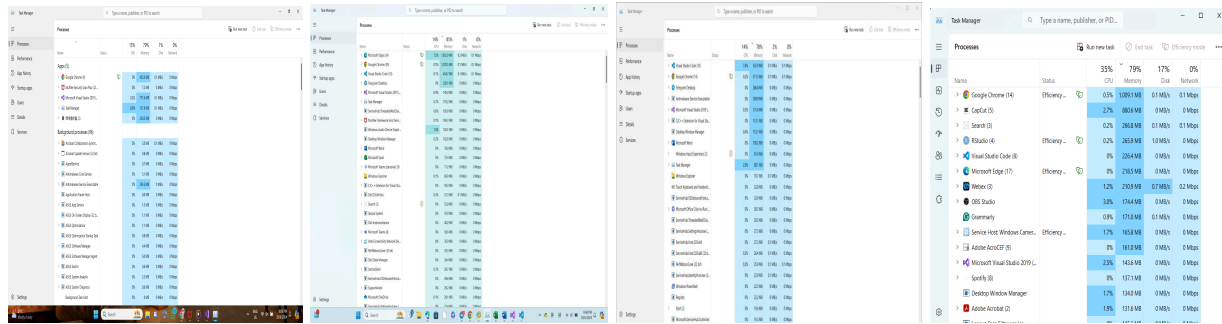
Middle Load Process



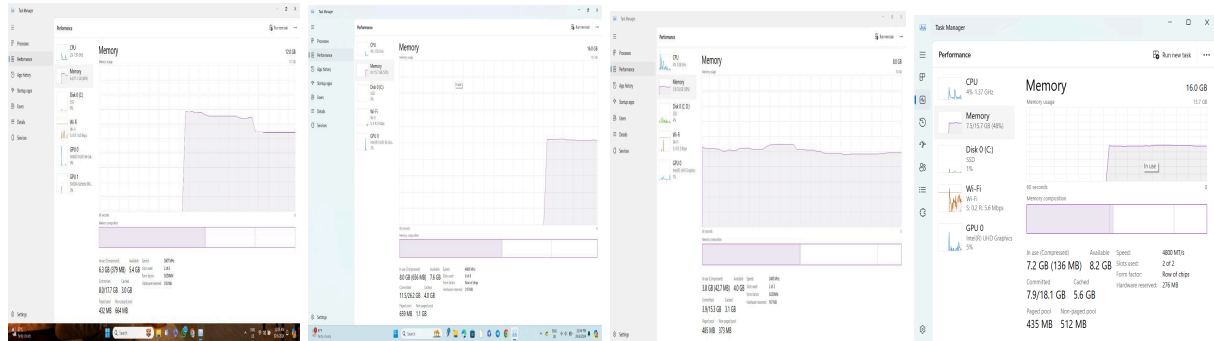
High Load Performance



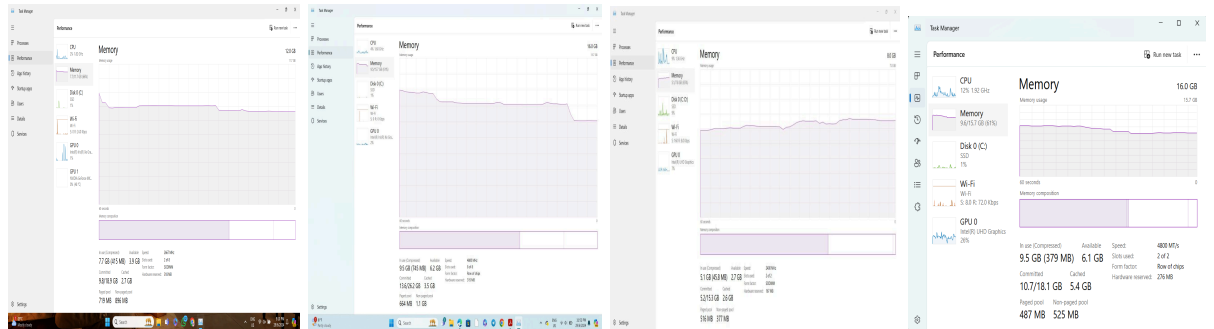
High Load Process



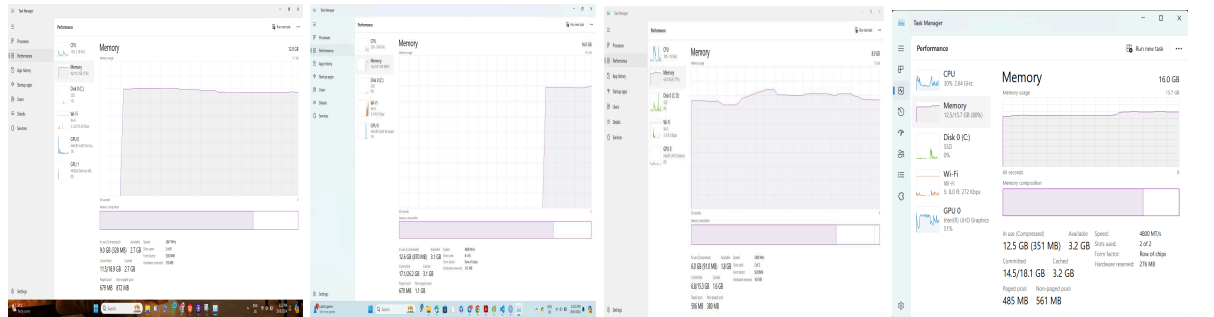
Low Load Memory



Middle Load Memory



High Load Memory



MiniTool Partition Wizard

Computer Type (Drive)	Performance				
	Transfer Size	Sequential Reading (MB/S)	Sequential Writing (MB/S)	Random Reading (MB/S)	Random Writing (MB/S)
Huawei Matebook D15 2022 (WDC PC SN530 SDBP)	32KB	1691.24	1730.75	2160.33	1483.59
	16KB	1422.64	1656.85	1542.2	1669.26
	8KB	1070.46	1219.79	892.72	894.117
	4KB	1049.34	613.147	880.414	571.479
Dell Inc. Inspiron 16 5630 (NVMe PC SN740 NVMe WD 1TB)	32KB	2584.36	1649.18	2141.33	1612.97
	16KB	1299.8	1043.77	1375.96	996.971
	8KB	1085.88	500.005	1320.44	602.912
	4KB	616.159	261.448	700.561	322.659
ASUS Vivobook 15 (NVMe INTEL SSDPEKNW 512GB RAID)	32KB	831.045	656.404	577.27	925.045
	16KB	596.992	904.393	485.414	61.1592
	8KB	483.149	545.087	332.631	506.001
	4KB	335.424	329.068	327.963	368.29
Lenovo 83BG (WD PC SN740 SDDPMQD-512G-1101)	32KB	1817.84	1726.3	2178.38	1745.32
	16KB	1158.44	881.96	1263.93	959.264
	8KB	945.031	579.126	1018.65	587.816
	4KB	513.06	280.058	532.482	303.617

MiniTool Partition Wizard Free 12.8

Data Backup | Data Recovery | Partition Recovery | Disk Benchmark | Space Analyzer

Bootable Media | Manual | Register

Partition Management | **Disk Benchmark**

Select a drive: C: Size 120.00 GB Free 15.42 GB Physical Disk: Disk 1 - WDC PC SN530 SDBPNPZ-512G-1027

Transfer Size: 4 KB to 32 KB Total Length: 200 MB

Queue Number: 8 Thread Number: 4 Test Mode: Sequential

Cool Down Time: 5 Seconds

Start

Drive C X

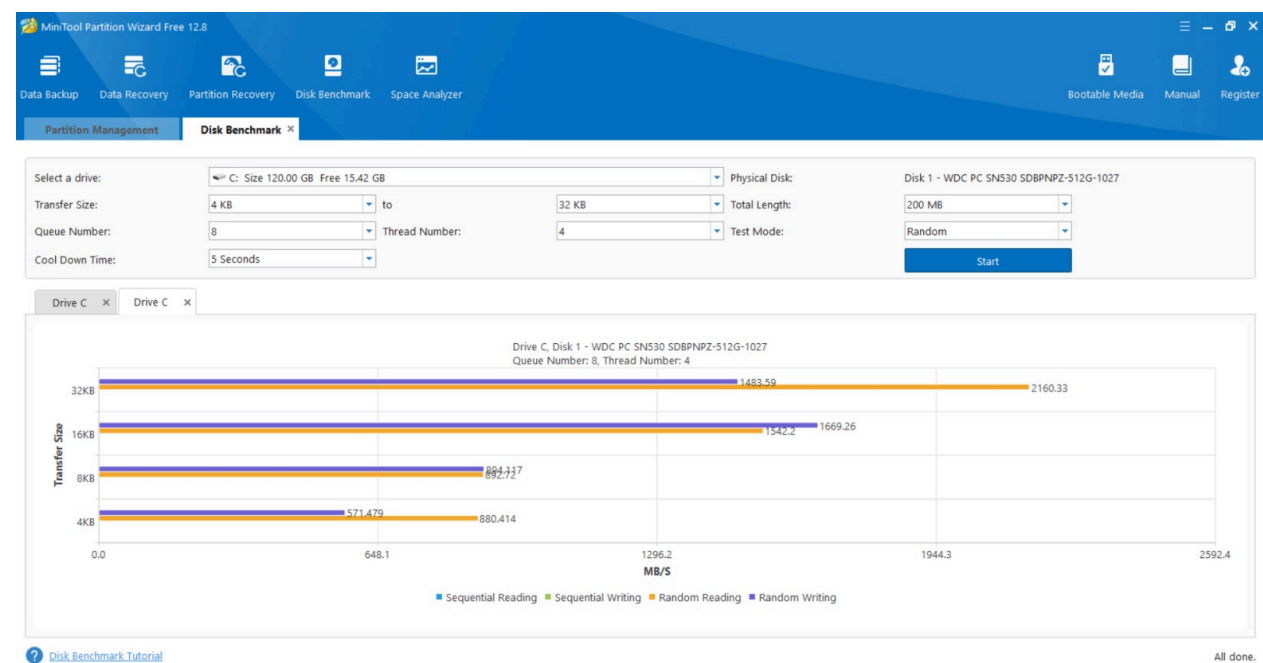
Drive C, Disk 1 - WDC PC SN530 SDBPNPZ-512G-1027
Queue Number: 8, Thread Number: 4

Transfer Size	Sequential Reading (MB/S)	Sequential Writing (MB/S)	Random Reading (MB/S)	Random Writing (MB/S)
32KB	1691.24	1730.75	-	-
16KB	1422.64	1656.85	-	-
8KB	1070.46	1219.79	-	-
4KB	1049.34	613.147	-	-

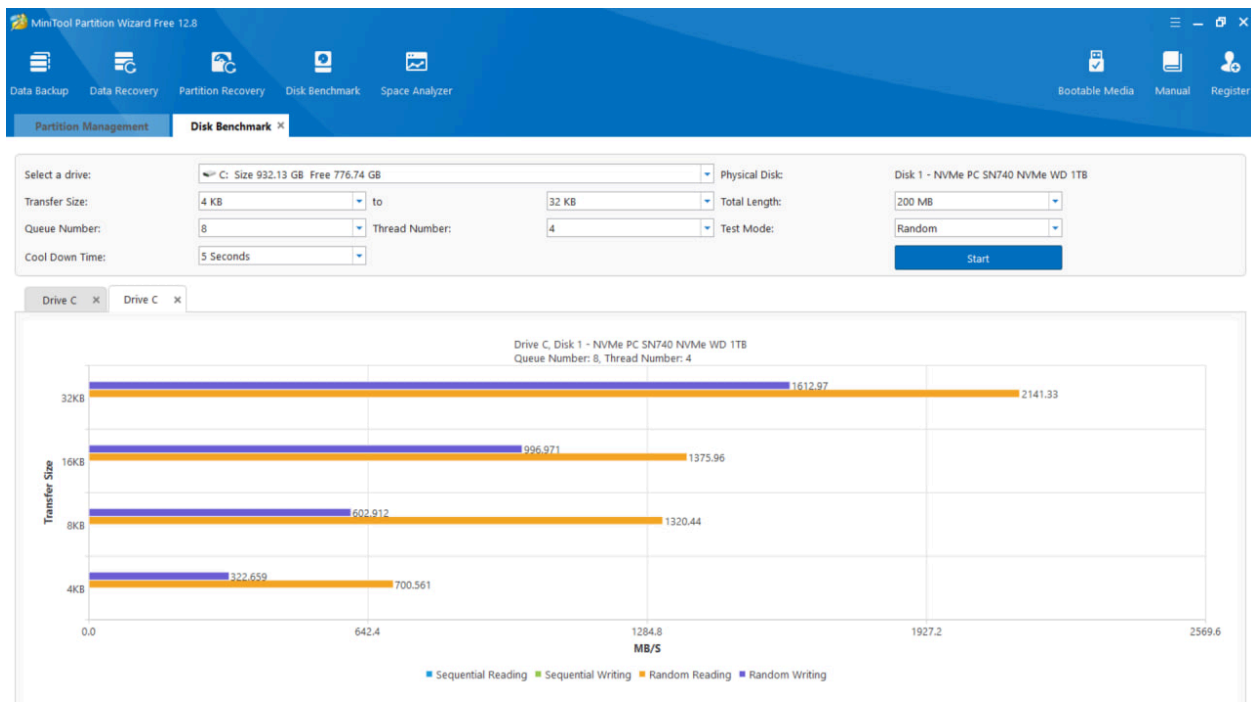
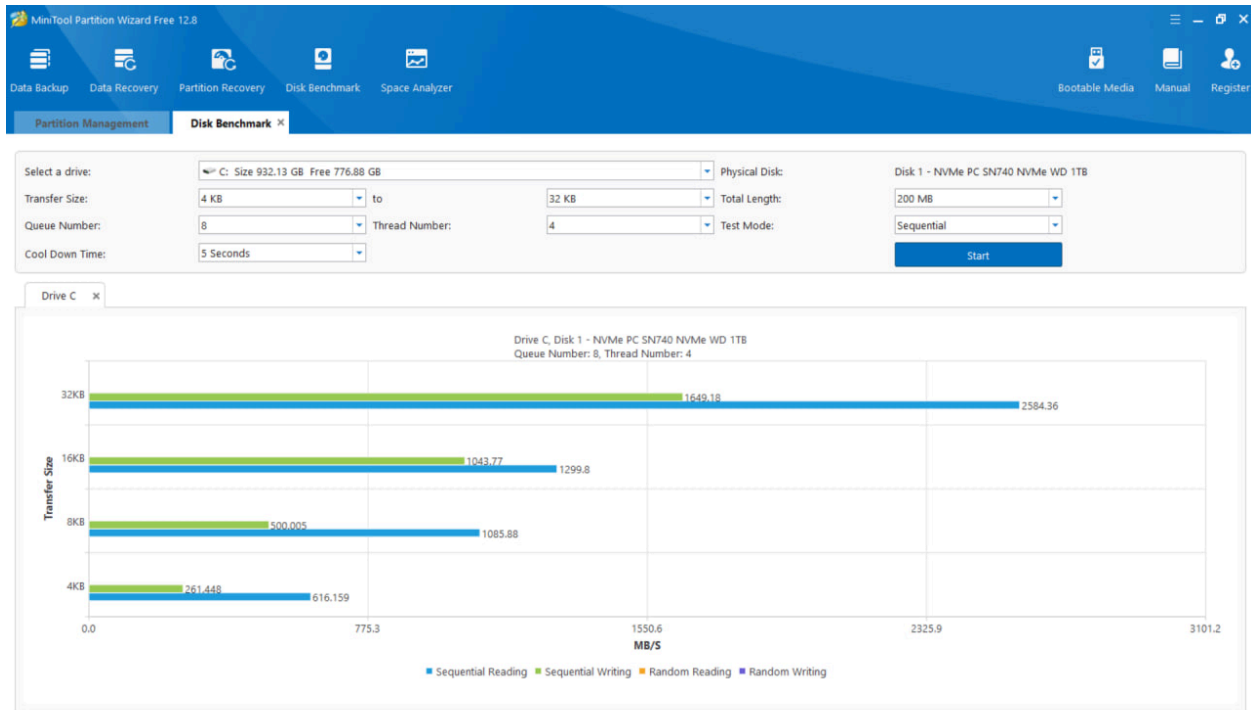
Legend: Sequential Reading (Blue), Sequential Writing (Green), Random Reading (Orange), Random Writing (Purple)

[Disk Benchmark Tutorial](#)

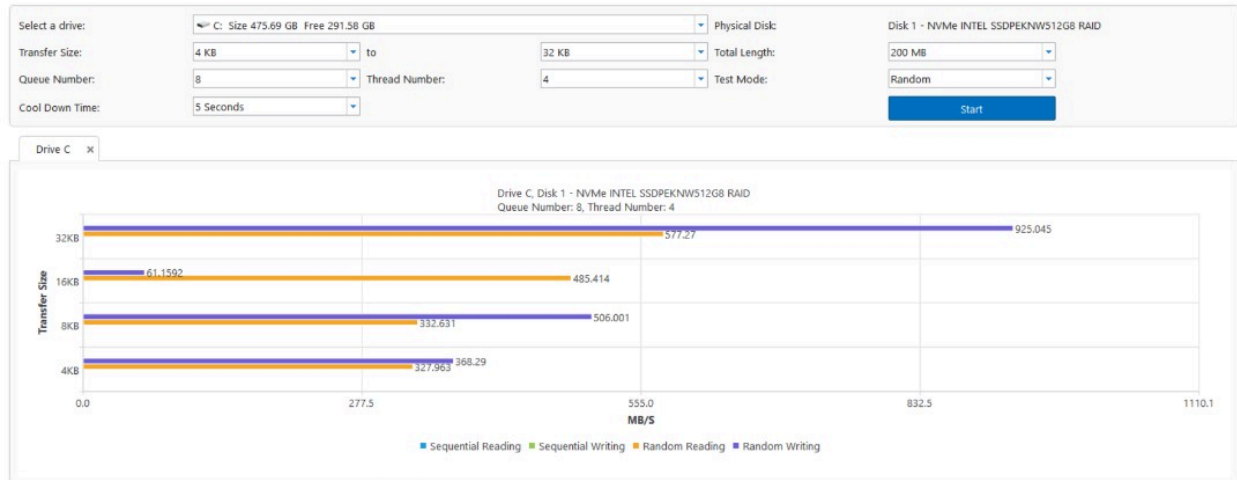
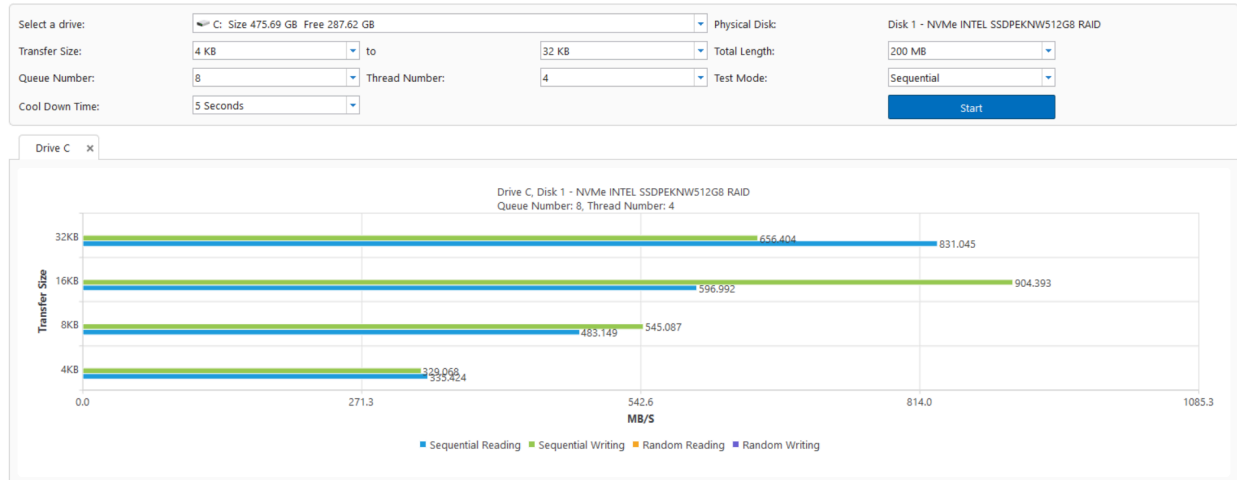
All done



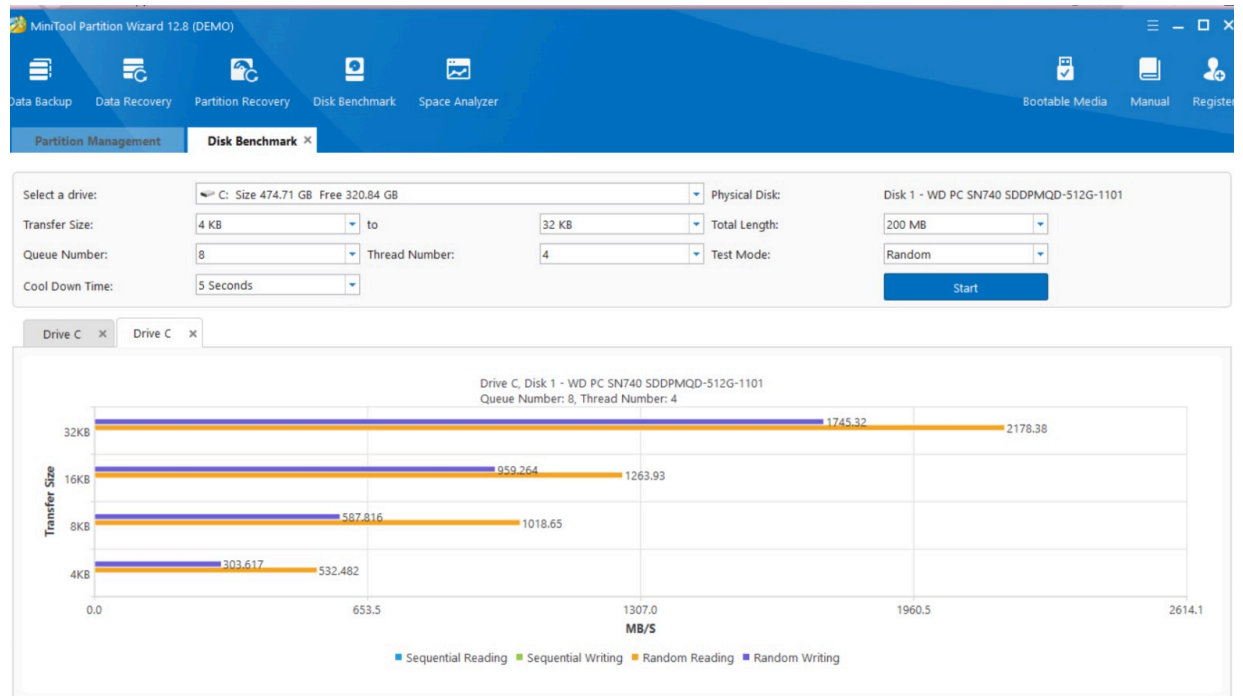
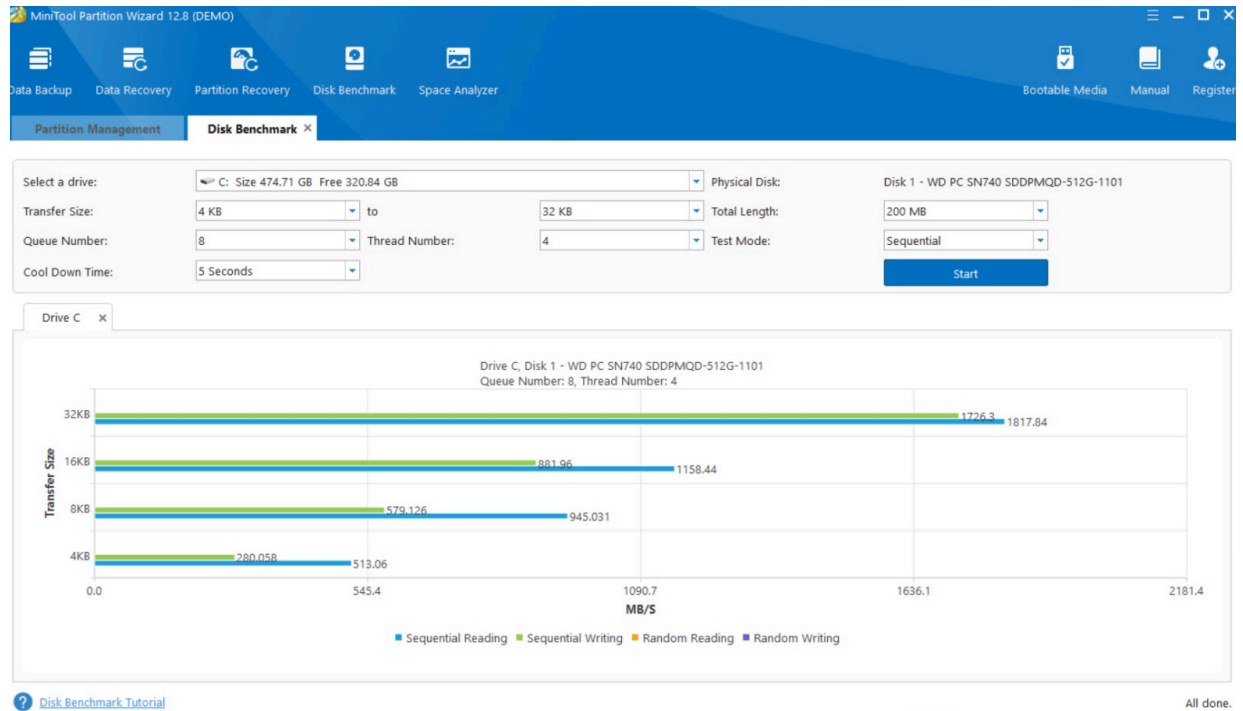
Dell Inc. Inspiron 16 5630 (NVMe PC SN740 NVMe WD 1TB)



ASUS Vivobook 15 (NVMe INTEL SSDPEKNW512GB RAID)



Lenovo 83BG (WD PC SN740 SDDPMQD-512G-1101)



2.4 Analysis, Comparison, Discussion

CPU-Z

1. CPU Technology

CPU technology involves creating the CPU with ever-smaller transistors, measured in nanometers (nm). Both the ASUS VivoBook 15, Dell Inspiron 16 5630, and Lenovo 83BG use a 10 nm process, which is more advanced and typically more power-efficient than the 14 nm process used in the Huawei MateBook D15 2022. Smaller process nodes (measured in nm) allow more transistors to be packed into the same physical space, increasing the computational power of the CPU. Smaller transistors generally consume less power, reducing the heat generated and improving battery life for mobile devices. Besides, smaller transistors can switch on and off faster, allowing for higher clock speeds, which can improve performance.

2. Number of Core

The ASUS VivoBook 15 and Huawei MateBook D15 2022 both have 4 cores, which is typical for many mainstream laptops but provides fewer simultaneous processing capabilities compared to the Dell Inspiron 16 5630 and Lenovo 83BG. The Dell Inspiron 16 5630 has the highest number of cores with 12, which allows it to handle more simultaneous processes and provides better performance for multi-threaded applications. The Lenovo 83BG has 8 cores, offering a balance between the core counts of the Dell Inspiron 16 5630 and the ASUS VivoBook 15/Huawei MateBook D15 2022. CPUs with more cores can handle more tasks simultaneously. Each core can process its own thread, allowing multiple threads to run in parallel. More cores can improve the overall responsiveness of a system, allowing it to run multiple applications at the same time without significant slowdowns.

3. RAM Size

RAM (Random Access Memory) is a crucial component that significantly impacts a laptop's performance and multitasking capabilities. More RAM enables better handling of multiple applications simultaneously and ensures smoother performance, particularly for memory-intensive tasks. The Dell Inspiron 16 5630 and Lenovo 83BG stand out with their 16 GB of RAM, offering superior performance for multitasking and demanding applications. The ASUS VivoBook 15 and Huawei MateBook D15 2022 both have 8 GB of RAM, which is sufficient for general use but less capable for heavy multitasking or professional-grade applications compared to the Dell Inspiron 16 5630 and Lenovo 83BG.

4. GPU Size

GPU memory, or VRAM (Video Random Access Memory), is dedicated memory used by the graphics processing unit (GPU) to store textures, frame buffers, and other graphical data. The size of the VRAM directly impacts the GPU's ability to handle complex textures and larger frame buffers. More VRAM enables the GPU to store more data, which is essential for high-resolution displays and modern games with detailed textures. For the ASUS VivoBook 15, the presence of the NVIDIA GeForce MX330 with 2 GB of GPU memory makes it the best choice for users needing better graphics performance. This includes gaming, video editing, and running graphics-heavy applications. The Dell Inspiron 16 5630, Huawei MateBook D15 2022, and Lenovo 83BG lack the additional power of a dedicated GPU, which becomes a limiting factor for graphics-intensive applications.

In conclusion, the **ASUS VivoBook 15**, with its dedicated NVIDIA GeForce MX330 GPU, is the best choice for graphics-intensive tasks such as gaming and video editing. The **Dell Inspiron 16 5630**, boasting a higher core count and 16 GB of RAM, excels in multitasking and professional applications. The **Lenovo 83BG**, with 8 cores and 16 GB of RAM, provides a good balance for both multitasking and general use but lacks a dedicated GPU for graphics-intensive tasks. For general use, both the **ASUS VivoBook 15** and **Huawei MateBook D15 2022** provide adequate performance, but the **ASUS VivoBook 15** offers an added advantage with its dedicated GPU. Therefore, the optimal choice depends on user needs: **ASUS VivoBook 15** for graphics-heavy tasks, **Dell Inspiron 16 5630** for multitasking and professional use, **Lenovo 83BG** for balanced performance, and **Huawei MateBook D15 2022** for general everyday tasks.

Geekbench

1. Add system info as complement /conformation to CPU-Z

For a more comprehensive understanding of the hardware configuration of each laptop, it is vital to combine system information with precise data from CPU-Z with this more detailed one from Geekbench. Key components in each laptop, like the processor (CPU), memory (RAM), cache memory, and chipset codename, play significant roles in determining performance and capabilities.

First of all, the **ASUS Vivobook 15** stands out with its Intel Core i5-1135G7 processor, which boasts 12 cores and 16 threads. This impressive configuration allows for exceptional performance in demanding applications. Operating at a peak frequency of 4193 MHz, the processor ensures swift and efficient handling of everyday tasks. The Tiger Lake chipset's cache memory sizes further enhance data processing speed. Additionally, this laptop comes with 12 GB of DDR4 SDRAM, operating at a transfer rate of 2660 MT/s over 2 channels, providing a balanced mix of speed and capacity for most users.

In contrast, the **Dell Inspiron 16** is equipped with a powerful Intel Core i7-1360P processor, featuring 4 cores and 8 threads. With a maximum frequency of 4987 MHz, this processor is designed to handle intensive workloads and multitasking with ease. Efficient distribution of tasks across clusters can enhance performance and lead to better load balancing, reducing bottlenecks. The Raptor Lake chipset's optimized cache memory sizes contribute to efficient data processing. This laptop is outfitted with 16 GB of DDR5 SDRAM, offering a high transfer rate of 4788 MT/s across 4 channels. This combination ensures superior performance for power users who need robust computing capabilities for tasks such as video editing, gaming, and professional applications.

On the other hand, the **Huawei Matebook** features an Intel Core i5-10210U processor, which includes 4 cores and 8 threads. The processor operates at a maximum frequency of 4189 MHz, delivering reliable performance for everyday computing needs such as web browsing, document editing, and multimedia consumption. The Comet Lake chipset's specific cache memory sizes improve data processing efficiency, ensuring smooth operation. This laptop comes with 8 GB of DDR4 SDRAM, which runs at a transfer rate of 2392 MT/s across 2 channels. This setup provides a dependable and efficient computing experience for users with standard requirements.

The **LENOVO 83BG** is equipped with an Intel Core i5-12450H processor, featuring two CPU clusters with 4 cores each. With a maximum operating frequency of 4389MHz, this processor, paired with the Comet Lake chipset, offers specific cache memory sizes that enhance data processing efficiency for seamless performance. Boasting 8 GB of DDR4 SDRAM across 2

channels running at a transfer rate of 2392 MT/s, this laptop provides a reliable and efficient computing experience for users with basic needs. Its well-optimized clusters and cache contribute to enhanced performance, stability, and efficiency.

In summary, combining system information with CPU-Z details allows for an in-depth understanding of each laptop's hardware components. By examining cache memory sizes, memory configurations, and chipset codenames (Tiger Lake, Raptor Lake, Comet Lake), you can make informed comparisons and choose the laptop that best meets your specific needs.

By comparing four of the computers, we could summarize that the most excellent was the **Dell Inspiron 16**. The computer cache and total cores of the CPU cluster can significantly influence performance of the laptops under a very heavy load. The **Dell Inspiron 16** has the largest and most efficient caches, they reduce the time the CPU needs to access data from the main memory, improving overall performance and reducing the CPU's workload, potentially lowering heat output. Overall, a well-optimized cache helps maintain performance and stability under stress. The total number of CPU clusters can also affect performance, particularly under heavy loads. More clusters can lead to better load balancing, reducing bottlenecks due to the efficient distribution of tasks across clusters.

2. CPU test with variation of technique or method used, either single-core or multi-core performance

A single-core method is best for tasks that rely on sequential processing, where each instruction must be completed before the next one starts or tasks that do not benefit from parallel processing, such as single-threaded applications and certain games. High single-core performance ensures these tasks run smoothly and quickly. The efficiency of a single core is measured by how quickly one core can complete a series of instructions. Here is an example of single core usage: a user who primarily browses the web, works with office applications, and plays older games might prioritize single-core performance.

While for a multi-core method, it can do tasks by splitting into multiple parallel processes, such as video rendering, scientific simulations, and running multiple applications simultaneously. Performance improvements come from adding more cores and optimizing how tasks are distributed among them. Higher scores indicate better performance in applications like web browsing and basic productivity software. High multi-core performance efficiently handles intensive, multitasking workloads. The efficiency of multi-core is measured by how effectively multiple cores can work together to complete parallel tasks. For instance, a user who edits

videos, performs complex data analysis or runs virtual machines would benefit more from high multi-core performance.

When comparing the **ASUS Vivobook**, **Dell Inspiron**, **Huawei Matebook** and **LENOVO 83BG** laptops. The **Dell Inspiron** stands out with the highest single-core score of 2336. In contrast, the **ASUS Vivobook** scores 1739, the **Huawei Matebook** scores 1207. And the **LENOVO** scores 1078 only. This indicates that the **Dell Inspiron 16** excels at tasks that rely heavily on single-core processing power.

Regarding multi-core performance, the **Dell Inspiron 16** also leads with a score of 9596, beating the **ASUS Vivobook** (4250), **Huawei Matebook** (3335) and the **LENOVO** (4653). This showcases the **Dell Inspiron 16** has a high ability to handle multiple applications simultaneously and efficiently manage tasks that benefit from parallel processing.

In conclusion, the **Dell Inspiron 16** boasts the most advanced and updated single and multi-core processing power compared to the other three laptops making it an excellent choice for users requiring robust processing capabilities. Nevertheless, the other three laptops might offer competitive performance in their respective categories too. The **ASUS Vivobook**, for instance, may appeal to those who prioritize battery life, portability, or budget considerations. Its respectable performance scores indicate it can handle everyday tasks efficiently, making it a solid option for students, office workers, and casual users who need a reliable machine without breaking the bank. While the **Huawei Matebook**, despite having the lowest performance scores among the three, might offer unique features such as a sleek design, a superior display quality, or additional security measures like a fingerprint scanner. Lastly, for the **LENOVO**, it offers remarkable durability and high-end security. They are ideal for businesses looking for a reliable laptop for stable and long-term use.

Task Manager

1. CPU Speed

Lenovo demonstrates the fastest speed at a high load, which is 2.58 GHz compared to Dell(1.53 GHz), Huawei(1.18 GHz) and ASUS(1.78 GHz). The fastest speed can provide better performance during high load. Thus, Lenovo is well suited for applications demanding high-speed computation, leading to superior performance during high load compared to other computers.

2. CPU Threads

Dell has the highest number of CPU threads, 5766 followed by Lenovo with 5117 threads, ASUS with 3585 threads and Huawei with 2947 threads. The higher number of CPU threads indicates Dell can manage more simultaneous tasks and improve multitasking performance.

3. Memory in use

For memory in use, Huawei has the least amount of memory used at a high load which is 6.0 GB compared to ASUS(9.0 GB), Dell(12.6 GB) and Lenovo(12.5 GB). It indicates that Huawei can manage its memory more effectively. This suggests that Huawei can handle tasks without excessive memory and perform better under high loads.

4. Memory committed

Dell has the highest memory committed at high load which is 17.1/26.2 GB compared to ASUS (11.5/18.6 GB), Huawei (6.8/15.3 GB) and Lenovo(14.5/18.1 GB). This indicates that Dell is more capable of handling more intensive tasks. This capability is important for applications that require large amounts of memory. Thus, Dell is suitable for multitasking.

5. Memory cached

Lenovo has the highest memory cached at every type of load, 5.6GB for low load, 5.4 GB for middle load and 3.2 GB for high load compared to ASUS (3.0 GB, 2.7 GB and 2.7 GB), Dell (4.0 GB, 3.5 GB and 3.1 GB) and Huawei (3.1 GB, 2.6 GB and 1.6 GB). Higher cached memory means better performance in accessing frequently used files and applications, improving system speed. Thus, Lenovo provides a smooth user experience that relies on quick access to cached data.

6. Memory paged pool

ASUS demonstrates the highest memory paged pool, 679 MB. This is higher compared to Dell(670 MB), Huawei(506 MB) and Lenovo(485 MB). This indicates that ASUS can manage its memory resources efficiently.

7. Memory non-paged pool

Dell has the highest memory non-paged pool at every type of load which is 1.1 GB. At high load for memory non-paged pool, ASUS has 872 MB, Huawei has 380 MB and Lenovo has 561 MB. This indicates that Dell can manage critical tasks more efficiently as it allocates more memory that is always available for crucial system operations and applications.

Analysis

The ASUS Vivobook 15 performs well in low and medium loads, with a high CPU speed and efficient memory consumption, although it has lower CPU utilization under high loads. It also has the highest memory paged pool indicating efficient memory management across various applications. This suggests that while ASUS may not handle intensive work efficiently, it is appropriate for ordinary chores.

Dell Inc.'s Inspiron 16 shows superior performance with the highest number of CPU threads, memory committed and non-paged at high load. This indicates that Dell can manage critical tasks more efficiently as it allocates more memory that is always available for crucial system operations and applications.

Huawei Matebook D15 performs moderately well with good efficiency in low and medium loads, but falls behind Dell in heavy loads, displaying lower CPU and memory utilization. This suggests that, while it manages daily tasks effectively, it may struggle with excessive multitasking, resulting in poor performance under high load.

Lenovo IdeaPad Slim 5 performs well in high loads with high CPU speed and the highest memory cached across all load levels. This makes it highly efficient for tasks requiring quick processing and frequent data access. However, it falls behind Dell in terms of CPU threads and memory committed. This indicates that Lenovo performs well in high-speed computations exceptionally.

MiniTool Partition Wizard

Analysis

This analysis compares the performance of four laptops with different SSDs using MiniTool Partition Wizard. The focus is on their sequential and random read/write capabilities across various block sizes.

The performance data from the four computers shows significant differences in their sequential and random read/write capabilities. The **Dell Inspiron 16** with NVMe PC SN740 is well-suited for tasks requiring high sequential read speeds, such as loading large files and operating system boot times. Its random read and write performance makes it strong for tasks involving frequent access to small files or database operations.

The **Lenovo 83BG** with WD PC SN740 demonstrates impressive sequential read performance at larger block sizes, making it effective for handling large file transfers and system booting. Its robust random read/write capabilities indicate it can handle tasks requiring frequent small file access, such as software development and database management.

The **Huawei Matebook D15** with WDC PC SN530 excels in sequential write tasks, making it suitable for activities like video editing and large file transfers. Its consistent random performance across different block sizes indicates reliability for general use and multitasking.

The **ASUS Vivobook 15** with NVMe INTEL SSDPEKNW shows weaker performance in both sequential and random operations, suggesting it might struggle with demanding applications. Its inconsistent performance, particularly in random writes, could lead to slower performance in everyday tasks involving frequent data access and modification.

1. Sequential Performance

Laptop (SSD)	Strengths	Weaknesses
Dell Inspiron 16 (NVMe PC SN740 NVMe WD 1TB)	The highest sequential reading speed at 32KB (2584.36 MB/s), a strong performance at larger block sizes.	Performance drops significantly at smaller block sizes.
Lenovo 83BG (WD PC SN740)	Strong sequential reading and writing performance, particularly at 32KB (1817.84 MB/s) and (1726.3 MB/s).	Noticeable performance drop at smaller block sizes (4KB).
Huawei Matebook D15 (WDC PC SN530)	Consistent performance across all block sizes, exceptional sequential writing at larger block sizes.	Less efficient for smaller block sizes.
ASUS Vivobook 15 (NVMe INTEL SSDPEKNW)	None particularly notable.	Weakest performance in both sequential reading and writing across all block sizes.

2. Random Performance

Laptop (SSD)	Strengths	Weaknesses
Dell Inspiron 16 (NVMe PC SN740 NVMe WD 1TB)	Strong performance in random reading and writing at larger block sizes.	Performance drops at smaller block sizes.
Lenovo 83BG (WD PC SN740)	Excels in random reading at larger block sizes (32KB: 2178.38 MB/s), strong random writing performance.	Decreases at smaller block sizes.
Huawei Matebook D15 (WDC PC SN530)	Good random reading performance, especially at larger block sizes.	Drops in random writing at smaller block sizes.
ASUS Vivobook 15 (NVMe INTEL SSDPEKNW)	None particularly notable.	Inconsistent and generally lower performance in random tasks.

Sequential Mode Stress Test

Based on the disk benchmarking results using MiniTool Partition Wizard, the Dell Inspiron 16, Lenovo 83BG, Huawei Matebook D15, and ASUS Vivobook 15 show varying performance in sequential read and write operations.

For sequential reading, the **Dell Inspiron 16** achieves the highest speeds, ranging from 616.159 MB/s to 2584.36 MB/s, making it the best performer at larger block sizes. The **Lenovo 83BG** also shows strong performance with speeds between 513.06 MB/s and 1817.84 MB/s, particularly effective at 32KB and 16KB transfer sizes. The **Huawei Matebook D15**

demonstrates consistent sequential read speeds across all block sizes, with results from 1049.34 MB/s to 1691.24 MB/s, indicating reliability for general use. In contrast, the **ASUS Vivobook 15** shows the weakest performance in sequential reading, achieving speeds from 335.424 MB/s to 831.045 MB/s, suggesting limitations in handling tasks requiring fast data access.

In sequential write operations, the **Dell Inspiron 16** again leads with speeds peaking at 1649.18 MB/s and maintaining strong performance at larger block sizes, though it drops significantly at smaller block sizes (down to 261.448 MB/s at 4KB). The **Lenovo 83BG** also performs well in sequential writing, with speeds reaching 1726.3 MB/s at 32KB and maintaining robust performance at 16KB and 8KB, though it drops at 4KB to 280.058 MB/s. The **Huawei Matebook D15** shows exceptional sequential writing speeds, particularly at 32KB (1730.75 MB/s) and 16KB (1656.85 MB/s), but it experiences a notable decrease at smaller block sizes (613.147 MB/s at 4KB). The **ASUS Vivobook 15** continues to show modest performance in sequential writing, with speeds between 329.068 MB/s and 656.404 MB/s, indicating potential limitations in handling tasks requiring rapid data transfer.

Overall, the **Dell Inspiron 16** demonstrates the highest throughput for both sequential read and write tasks, making it suitable for applications requiring fast access to large files. The **Lenovo 83BG** also offers strong all-around performance, especially effective for large file transfers and system booting. The **Huawei Matebook D15** shows reliable and consistent sequential performance, particularly excelling in sequential writing at larger block sizes. In contrast, the **ASUS Vivobook 15** exhibits the weakest performance, indicating it may struggle with more demanding applications and is better suited for less intensive tasks.

Random Mode Stress Test

For random reading, the **Dell Inspiron 16** achieves strong performance, with speeds ranging from 700.561 MB/s at 4KB to 2141.33 MB/s at 32KB, making it well-suited for tasks involving frequent access to small files. The **Lenovo 83BG** also excels in random read, with speeds from 532.482 MB/s at 4KB to 2178.38 MB/s at 32KB, indicating robust performance across different block sizes. The **Huawei Matebook D15** demonstrates good random read speeds, especially at larger block sizes, with results from 880.414 MB/s at 4KB to 2160.33 MB/s at 32KB, making it a solid choice for multitasking and general use. In contrast, the **ASUS Vivobook 15** shows the weakest random read performance, achieving speeds from 327.963 MB/s at 4KB to 577.27 MB/s at 32KB, suggesting limitations in handling tasks requiring quick data access.

In random write operations, the **Dell Inspiron 16** maintains strong performance, with speeds peaking at 1612.97 MB/s at 32KB and remaining robust at 16KB and 8KB, though it drops to 322.659 MB/s at 4 KB. The **Lenovo 83BG** also performs well in random writing, with speeds reaching 1745.32 MB/s at 32 KB and maintaining solid performance at 16KB and 8KB, though

it drops at 4KB to 303.617 MB/s. The **Huawei Matebook D15** shows commendable random write speeds, particularly at larger block sizes, with results from 571.479 MB/s at 4KB to 1483.59 MB/s at 32KB, indicating reliability for general use and multitasking. The **ASUS Vivobook 15** continues to show modest performance in random writing, with speeds between 368.29 MB/s at 4KB and 925.045 MB/s at 32KB, indicating potential limitations in handling tasks requiring frequent data modification.

Overall, the **Dell Inspiron 16** and **Lenovo 83BG** offer the best all-around performance in random read and write tasks, making them ideal for high-performance tasks involving frequent access and modification of small files. The **Huawei Matebook D15** shows reliable and consistent random performance, particularly excelling in random writing at larger block sizes, making it a solid choice for multitasking and general use. In contrast, the **ASUS Vivobook 15** exhibits the weakest performance in both random read and write tasks, indicating it may struggle with more demanding applications and is better suited for less intensive tasks.

Conclusion

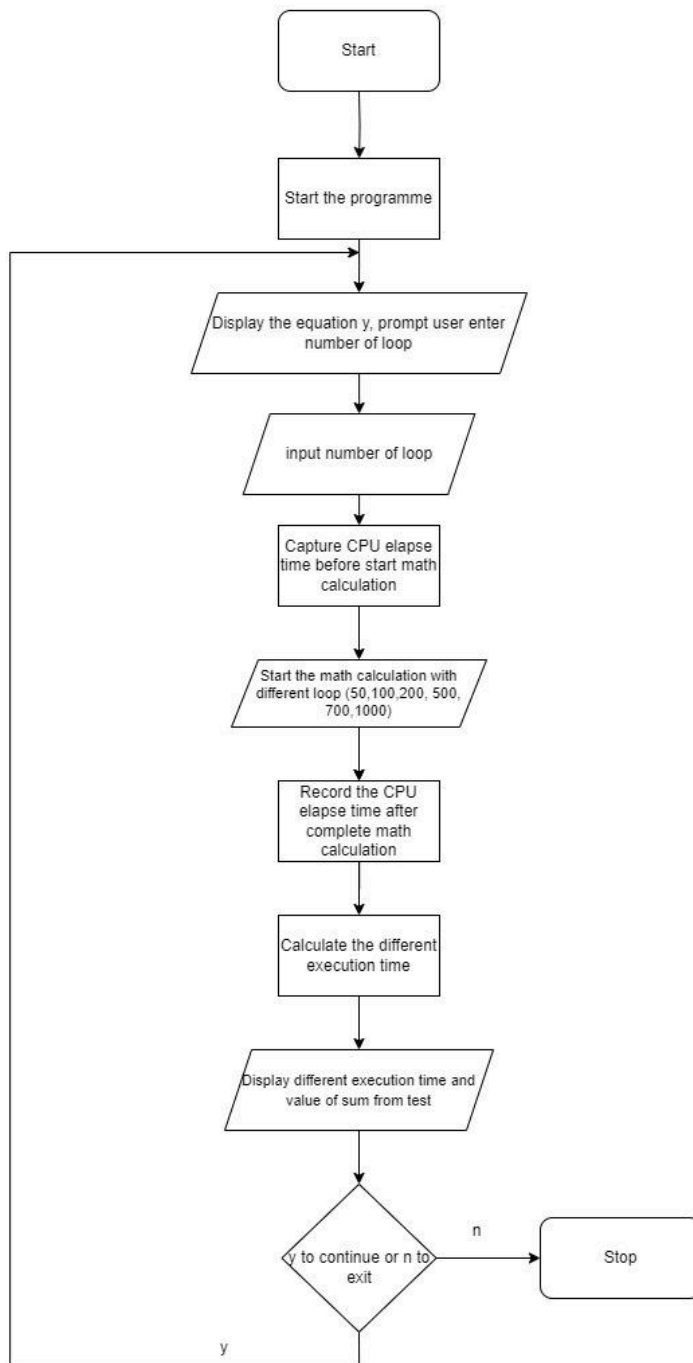
In summary, the **Dell Inspiron 16 with NVMe PC SN740** and the **Lenovo 83BG with WD PC SN740** offer the best all-around performance, excelling in both sequential and random tasks. It is well-suited for high-performance tasks that require fast data access and handling of large files. The Huawei Matebook D15 with WDC PC SN530 shows strong sequential writing performance and consistent random performance, making it a reliable choice for both large file operations and general multitasking. In contrast, the ASUS Vivobook 15 with NVMe INTEL SSDPEKNW exhibits the weakest performance across most metrics, indicating it may be less suitable for tasks requiring high disk performance and may only suffice for less demanding applications.

3.0 Project - Part 2

3.1 Overview

In this part of the project, we aim to investigate the CPU elapsed time for four different computers. The task involves writing an assembly language program, calculating the running time before execution, and then recording the actual time it takes for the program to run. After obtaining the recorded times, we will employ the provided mathematical equations to measure the efficiency, speed, and capability of each CPU. These results will be analyzed, compared, and discussed to understand the performance differences between the systems.

3.2 Flowchart of Execution



3.3 Coding and Implementation

TITLE Project COA

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INCLUDE Irvine32.inc

.data

prompt1 **BYTE** "Welcome to CPU Benchmark Program", 0Dh, 0Ah, 0
prompt2 **BYTE** "Benchmark CPU time Using Equation $y = 2*x^3 + 14*x^2 + 18*x + 22$ ", 0Dh, 0Ah,
"(with delay coef1,coef2,coef3,coef4 = 2,14,18,22 msec)", 0Dh, 0Ah, 0
prompt4 **BYTE** "Enter Number of Looping (N) = ", 0
prompt5 **BYTE** "CPU time Stress Test in progress...", 0Dh, 0Ah, 0
prompt6 **BYTE** "Result:", 0Dh, 0Ah, 0
prompt7 **BYTE** "First Capture Execution time in millisecond: ", 0
prompt8 **BYTE** "Second Capture Execution time in millisecond: ", 0
prompt9 **BYTE** "Different Execution time in millisecond: ", 0
prompt10 **BYTE** "Value of Sum from the Stress Test (polynomial) = ", 0
prompt11 **BYTE** "Press 'y' to continue or 'n' to exit the benchmark: ", 0
msgBye **BYTE** "Thank you... BYE!!", 0DH, 0AH, 0

coef1 **DWORD** 2
coef2 **DWORD** 14
coef3 **DWORD** 18
coef4 **DWORD** 22

N **DWORD** ? ; number of loops
x **DWORD** 1
sum **DWORD** 0
start_time **DWORD** ?
end_time **DWORD** ?
elapsed_time **DWORD** ?

.code

main **PROC**

; Display welcome message

call Clrscr

```

mov edx, OFFSET prompt1
call WriteString
call CrLf

; Display the algorithm
mov edx, OFFSET prompt2
call WriteString
call CrLf

; Prompt for max loop value
mov edx, OFFSET prompt4
call WriteString
call ReadDec
mov N, eax

; Display progress message
mov edx, OFFSET prompt5
call WriteString
call CrLf

; Capture time before starting the loop
call GetMseconds
mov start_time, eax

; Initialize sum to 0
mov sum, 0

mov ecx, N; Set loop counter
mov ebx, 1; Initialize ebx to 1

```

calc_loop:

```

; Calculate  $y = (\text{coef1} * x^3) + (\text{coef2} * x^2) + (\text{coef3} * x) + \text{coef4}$ 

;  $\text{coef1} * x^3$ 
mov eax, coef1
call Delay ; Introduce a delay
imul eax, x ; Multiply eax by x
imul eax, x ; Multiply eax by x again( $\text{eax} = 2 * x^2$ )
imul eax, x ; Multiply eax by x again( $\text{eax} = 2 * x^3$ )
mov edx, eax ; Move the result into edx( $\text{edx} = 2 * x^3$ )

```

```

; coef2* x ^ 2
mov eax, coef2
call Delay
imul eax, x
imul eax, x
add edx, eax          ; Add eax to edx( $edx = 2 * x^3 + 14 * x^2$ )

; coef3* x
mov eax, coef3
call Delay
imul eax, x
add edx, eax          ; add previous result

; coef4
add edx, coef4        ; Add coef4(22) to edx( $edx = 2 * x^3 + 14 * x^2 + 18 * x + 22$ )

; Add y to sum
add sum, edx

; Increment loop counter
inc x                  ; Increment x by 1
loop calc_loop        ; Decrement ecx by 1 and loop if ecx != 0

; Capture time after completing the loop
call GetMseconds
mov end_time, eax

; Calculate elapsed time
mov eax, end_time
sub eax, start_time
mov elapsed_time, eax

; Display results
mov edx, OFFSET prompt6
call WriteString
call Crlf

mov edx, OFFSET prompt7

```

```

    call WriteString
    mov eax, start_time
    call WriteDec
    call Crlf

    mov edx, OFFSET prompt8
    call WriteString
    mov eax, end_time
    call WriteDec
    call Crlf

    mov edx, OFFSET prompt9
    call WriteString
    mov eax, elapsed_time
    call WriteDec
    call Crlf

    mov edx, OFFSET prompt10
    call WriteString
    mov eax, sum
    call WriteDec
    call Crlf

; Prompt to continue or exit
    mov edx, OFFSET prompt11
    call WriteString
    call ReadChar

    cmp al, 'y'
    je main

    mov edx, offset msgBye
    call WriteString

exit
main ENDP
END main

```

1. ASUS VIVOBOOK 15

No of Loop (N)	Capture_msec (before)	Capture_msec (after)	CPU_msec (different)
50	81787887	81720979	3092
100	81824238	81830440	6202
200	81926287	81938694	12407
500	81985726	82016822	31096
700	82079186	82122584	43398
1000	82277459	82339460	62001

```

C:\Users\User\source\repos\C > + v
Welcome to CPU Benchmark Program

Benchmark CPU time Using Equation  $y = 2x^3 + 14x^2 + 18x + 22$ 
(with delay coef1,coef2,coef3,coef4 = 2,14,18,22 msec)

Enter Number of Looping (N) = 50
CPU time Stress Test in progress...

Result:
First Capture Execution time in millisecond: 81717887
Second Capture Execution time in millisecond: 81720979
Different Execution time in millisecond: 3092
Value of Sum from the Stress Test (polynomial) = 3876250
Press 'y' to continue or 'n' to exit the benchmark: |

```

```

C:\Users\User\source\repos\C > + v
Welcome to CPU Benchmark Program

Benchmark CPU time Using Equation  $y = 2x^3 + 14x^2 + 18x + 22$ 
(with delay coef1,coef2,coef3,coef4 = 2,14,18,22 msec)

Enter Number of Looping (N) = 100
CPU time Stress Test in progress...

Result:
First Capture Execution time in millisecond: 81824238
Second Capture Execution time in millisecond: 81830440
Different Execution time in millisecond: 6202
Value of Sum from the Stress Test (polynomial) = 268750000
Press 'y' to continue or 'n' to exit the benchmark:

```

```

C:\Users\User\source\repos\C > + v
Welcome to CPU Benchmark Program

Benchmark CPU time Using Equation  $y = 2x^3 + 14x^2 + 18x + 22$ 
(with delay coef1,coef2,coef3,coef4 = 2,14,18,22 msec)

Enter Number of Looping (N) = 200
CPU time Stress Test in progress...

Result:
First Capture Execution time in millisecond: 81926287
Second Capture Execution time in millisecond: 81938694
Different Execution time in millisecond: 12407
Value of Sum from the Stress Test (polynomial) = 3180522704
Press 'y' to continue or 'n' to exit the benchmark:

```

```

C:\Users\User\source\repos\C > + v
Welcome to CPU Benchmark Program

Benchmark CPU time Using Equation  $y = 2x^3 + 14x^2 + 18x + 22$ 
(with delay coef1,coef2,coef3,coef4 = 2,14,18,22 msec)

Enter Number of Looping (N) = 500
CPU time Stress Test in progress...

Result:
First Capture Execution time in millisecond: 81985726
Second Capture Execution time in millisecond: 82016822
Different Execution time in millisecond: 31096
Value of Sum from the Stress Test (polynomial) = 3343929536
Press 'y' to continue or 'n' to exit the benchmark:

```

```

C:\Users\User\source\repos\C > + v
Welcome to CPU Benchmark Program

Benchmark CPU time Using Equation  $y = 2x^3 + 14x^2 + 18x + 22$ 
(with delay coef1,coef2,coef3,coef4 = 2,14,18,22 msec)

Enter Number of Looping (N) = 700
CPU time Stress Test in progress...

Result:
First Capture Execution time in millisecond: 82079186
Second Capture Execution time in millisecond: 82122584
Different Execution time in millisecond: 43398
Value of Sum from the Stress Test (polynomial) = 1244772960
Press 'y' to continue or 'n' to exit the benchmark:

```

```

C:\Users\User\source\repos\C > + v
Welcome to CPU Benchmark Program

Benchmark CPU time Using Equation  $y = 2x^3 + 14x^2 + 18x + 22$ 
(with delay coef1,coef2,coef3,coef4 = 2,14,18,22 msec)

Enter Number of Looping (N) = 1000
CPU time Stress Test in progress...

Result:
First Capture Execution time in millisecond: 82277459
Second Capture Execution time in millisecond: 82339460
Different Execution time in millisecond: 62001
Value of Sum from the Stress Test (polynomial) = 1551397968
Press 'y' to continue or 'n' to exit the benchmark:

```

2. Dell Inc. Inspiron 16

No of Loop (N)	Capture_msec (before)	Capture_msec (after)	CPU_msec (different)
50	876944	880043	3099
100	980966	987159	6193
200	1098314	1110763	12449
500	1132522	1164523	32001
700	1178757	1224343	45576
1000	1249419	1313813	64394

```

C:\Users\yeewe\source\repos x + v
Welcome to CPU Benchmark Program
Benchmark CPU time Using Equation y = 2*x^3 + 14*x^2 + 18*x + 22
(with delay coef1,coef2,coef3,coef4 = 2,14,18,22 msec)

Enter Number of Looping (N) = 50
CPU time Stress Test in progress...

Result:
First Capture Execution time in millisecond: 1415537
Second Capture Execution time in millisecond: 1418704
Different Execution time in millisecond: 3167
Value of Sum from the Stress Test (polynomial) = 519047954
Press 'y' to continue or 'n' to exit the benchmark: |

```

```

C:\Users\yeewe\source\repos x + v
Welcome to CPU Benchmark Program
Benchmark CPU time Using Equation y = 2*x^3 + 14*x^2 + 18*x + 22
(with delay coef1,coef2,coef3,coef4 = 2,14,18,22 msec)

Enter Number of Looping (N) = 100
CPU time Stress Test in progress...

Result:
First Capture Execution time in millisecond: 980966
Second Capture Execution time in millisecond: 987159
Different Execution time in millisecond: 6193
Value of Sum from the Stress Test (polynomial) = 55835000
Press 'y' to continue or 'n' to exit the benchmark: |

```

```

C:\Users\yeewe\source\repos x + v
Welcome to CPU Benchmark Program
Benchmark CPU time Using Equation y = 2*x^3 + 14*x^2 + 18*x + 22
(with delay coef1,coef2,coef3,coef4 = 2,14,18,22 msec)

Enter Number of Looping (N) = 200
CPU time Stress Test in progress...

Result:
First Capture Execution time in millisecond: 1098314
Second Capture Execution time in millisecond: 1110763
Different Execution time in millisecond: 12449
Value of Sum from the Stress Test (polynomial) = 4148660000
Press 'y' to continue or 'n' to exit the benchmark: |

```

```

C:\Users\yeewe\source\repos x + v
Welcome to CPU Benchmark Program
Benchmark CPU time Using Equation y = 2*x^3 + 14*x^2 + 18*x + 22
(with delay coef1,coef2,coef3,coef4 = 2,14,18,22 msec)

Enter Number of Looping (N) = 500
CPU time Stress Test in progress...

Result:
First Capture Execution time in millisecond: 1132522
Second Capture Execution time in millisecond: 1164523
Different Execution time in millisecond: 32001
Value of Sum from the Stress Test (polynomial) = 1643962088
Press 'y' to continue or 'n' to exit the benchmark: |

```

```

C:\Users\yeewe\source\repos x + v
Welcome to CPU Benchmark Program
Benchmark CPU time Using Equation y = 2*x^3 + 14*x^2 + 18*x + 22
(with delay coef1,coef2,coef3,coef4 = 2,14,18,22 msec)

Enter Number of Looping (N) = 700
CPU time Stress Test in progress...

Result:
First Capture Execution time in millisecond: 1178757
Second Capture Execution time in millisecond: 1224343
Different Execution time in millisecond: 45576
Value of Sum from the Stress Test (polynomial) = 1943078680
Press 'y' to continue or 'n' to exit the benchmark: |

```

```

C:\Users\yeewe\source\repos x + v
Welcome to CPU Benchmark Program
Benchmark CPU time Using Equation y = 2*x^3 + 14*x^2 + 18*x + 22
(with delay coef1,coef2,coef3,coef4 = 2,14,18,22 msec)

Enter Number of Looping (N) = 1000
CPU time Stress Test in progress...

Result:
First Capture Execution time in millisecond: 1249419
Second Capture Execution time in millisecond: 1313813
Different Execution time in millisecond: 64394
Value of Sum from the Stress Test (polynomial) = 1282665696
Press 'y' to continue or 'n' to exit the benchmark: |

```

3. Huawei Matebook D15 2022

No of Loop (N)	Capture_msec (before)	Capture_msec (after)	CPU_msec (different)
50	84083467	84086659	3192
100	84178055	84184393	6338
200	84236410	84248397	11987
500	84290928	84321754	30826
700	84666038	84707456	41418
1000	84768773	84828960	60187

```

C:\Users\braya\source\repos\ X + v
Benchmark CPU time Using Equation  $y = 2x^3 + 14x^2 + 18x + 22$ 
(with delay coef1,coef2,coef3,coef4 = 2,14,18,22 msec)

Enter Number of Looping (N) = 50
CPU time Stress Test in progress...

Result:

First Capture Execution time in millisecond: 84083467
Second Capture Execution time in millisecond: 84086659
Different Execution time in millisecond: 3192
Value of Sum from the Stress Test (polynomial) = 3876250
Press 'y' to continue or 'n' to exit the benchmark: |

```

```

C:\Users\braya\source\repos\ X + v
Welcome to CPU Benchmark Program

Benchmark CPU time Using Equation  $y = 2x^3 + 14x^2 + 18x + 22$ 
(with delay coef1,coef2,coef3,coef4 = 2,14,18,22 msec)

Enter Number of Looping (N) = 100
CPU time Stress Test in progress...

Result:

First Capture Execution time in millisecond: 84178055
Second Capture Execution time in millisecond: 84184393
Different Execution time in millisecond: 6338
Value of Sum from the Stress Test (polynomial) = 268750000
Press 'y' to continue or 'n' to exit the benchmark: |

```

```

C:\Users\braya\source\repos\ X + v
Welcome to CPU Benchmark Program

Benchmark CPU time Using Equation  $y = 2x^3 + 14x^2 + 18x + 22$ 
(with delay coef1,coef2,coef3,coef4 = 2,14,18,22 msec)

Enter Number of Looping (N) = 200
CPU time Stress Test in progress...

Result:

First Capture Execution time in millisecond: 84236410
Second Capture Execution time in millisecond: 84248397
Different Execution time in millisecond: 11987
Value of Sum from the Stress Test (polynomial) = 3180522704
Press 'y' to continue or 'n' to exit the benchmark: |

```

```

C:\Users\braya\source\repos\ X + v
Welcome to CPU Benchmark Program

Benchmark CPU time Using Equation  $y = 2x^3 + 14x^2 + 18x + 22$ 
(with delay coef1,coef2,coef3,coef4 = 2,14,18,22 msec)

Enter Number of Looping (N) = 500
CPU time Stress Test in progress...

Result:

First Capture Execution time in millisecond: 84290928
Second Capture Execution time in millisecond: 84321754
Different Execution time in millisecond: 30826
Value of Sum from the Stress Test (polynomial) = 3343929536
Press 'y' to continue or 'n' to exit the benchmark: |

```

```

C:\Users\braya\source\repos\ X + v
Welcome to CPU Benchmark Program

Benchmark CPU time Using Equation  $y = 2x^3 + 14x^2 + 18x + 22$ 
(with delay coef1,coef2,coef3,coef4 = 2,14,18,22 msec)

Enter Number of Looping (N) = 700
CPU time Stress Test in progress...

Result:

First Capture Execution time in millisecond: 84666038
Second Capture Execution time in millisecond: 84707456
Different Execution time in millisecond: 41418
Value of Sum from the Stress Test (polynomial) = 1244772960
Press 'y' to continue or 'n' to exit the benchmark: |

```

```

C:\Users\braya\source\repos\ X + v
Welcome to CPU Benchmark Program

Benchmark CPU time Using Equation  $y = 2x^3 + 14x^2 + 18x + 22$ 
(with delay coef1,coef2,coef3,coef4 = 2,14,18,22 msec)

Enter Number of Looping (N) = 1000
CPU time Stress Test in progress...

Result:

First Capture Execution time in millisecond: 84768773
Second Capture Execution time in millisecond: 84828960
Different Execution time in millisecond: 60187
Value of Sum from the Stress Test (polynomial) = 1551397968
Press 'y' to continue or 'n' to exit the benchmark: |

```

4. Lenovo IdeaPad Slim 5

No of Loop (N)	Capture_msec (before)	Capture_msec (after)	CPU_msec (different)
50	85270505	85273854	3165
100	85291125	85297778	6457
200	85311266	85324450	12938
500	85337826	85370696	32364
700	85510241	85556318	45364
1000	85571398	85637804	64500

```

C:\Users\Lenovo\source\repo x + v
Welcome to CPU Benchmark Program

Benchmark CPU time Using Equation  $y = 2x^3 + 14x^2 + 18x + 22$ 
(with delay coef1,coef2,coef3,coef4 = 2,14,18,22 msec)

Enter Number of Looping (N) = 50
CPU time Stress Test in progress...

Result:
First Capture Execution time in millisecond: 38062213
Second Capture Execution time in millisecond: 38065378
Different Execution time in millisecond: 3165
Value of Sum from the Stress Test (polynomial) = 2161788750
Press 'y' to continue or 'n' to exit the benchmark: |

```

```

C:\Users\Lenovo\source\repo x + v
Welcome to CPU Benchmark Program

Benchmark CPU time Using Equation  $y = 2x^3 + 14x^2 + 18x + 22$ 
(with delay coef1,coef2,coef3,coef4 = 2,14,18,22 msec)

Enter Number of Looping (N) = 100
CPU time Stress Test in progress...

Result:
First Capture Execution time in millisecond: 39893508
Second Capture Execution time in millisecond: 39899965
Different Execution time in millisecond: 6457
Value of Sum from the Stress Test (polynomial) = 66183320
Press 'y' to continue or 'n' to exit the benchmark: |

```

```

C:\Users\Lenovo\source\repo x + v
Welcome to CPU Benchmark Program

Benchmark CPU time Using Equation  $y = 2x^3 + 14x^2 + 18x + 22$ 
(with delay coef1,coef2,coef3,coef4 = 2,14,18,22 msec)

Enter Number of Looping (N) = 200
CPU time Stress Test in progress...

Result:
First Capture Execution time in millisecond: 39277223
Second Capture Execution time in millisecond: 39290161
Different Execution time in millisecond: 12938
Value of Sum from the Stress Test (polynomial) = 3703387472
Press 'y' to continue or 'n' to exit the benchmark: |

```

```

C:\Users\Lenovo\source\repo x + v
Welcome to CPU Benchmark Program

Benchmark CPU time Using Equation  $y = 2x^3 + 14x^2 + 18x + 22$ 
(with delay coef1,coef2,coef3,coef4 = 2,14,18,22 msec)

Enter Number of Looping (N) = 500
CPU time Stress Test in progress...

Result:
First Capture Execution time in millisecond: 39135547
Second Capture Execution time in millisecond: 39167911
Different Execution time in millisecond: 32364
Value of Sum from the Stress Test (polynomial) = 4120395280
Press 'y' to continue or 'n' to exit the benchmark: |

```

```

C:\Users\Lenovo\source\repo x + v
Welcome to CPU Benchmark Program

Benchmark CPU time Using Equation  $y = 2x^3 + 14x^2 + 18x + 22$ 
(with delay coef1,coef2,coef3,coef4 = 2,14,18,22 msec)

Enter Number of Looping (N) = 700
CPU time Stress Test in progress...

Result:
First Capture Execution time in millisecond: 39469790
Second Capture Execution time in millisecond: 39515154
Different Execution time in millisecond: 45364
Value of Sum from the Stress Test (polynomial) = 4169962592
Press 'y' to continue or 'n' to exit the benchmark: |

```

```

C:\Users\Lenovo\source\repo x + v
Welcome to CPU Benchmark Program

Benchmark CPU time Using Equation  $y = 2x^3 + 14x^2 + 18x + 22$ 
(with delay coef1,coef2,coef3,coef4 = 2,14,18,22 msec)

Enter Number of Looping (N) = 1000
CPU time Stress Test in progress...

Result:
First Capture Execution time in millisecond: 38759533
Second Capture Execution time in millisecond: 38824033
Different Execution time in millisecond: 64500
Value of Sum from the Stress Test (polynomial) = 1174755216
Press 'y' to continue or 'n' to exit the benchmark: |

```


3.4 Analysis, Comparison, Discussion

CPU consumption time is affected by many factors, which directly show the performance of a computer. The factors that affect the running time of the CPU include hardware differences, memory differences, operating system and software, background processes, and system load.

After writing the assembly program completely, we ran it on four different computers to capture the CPU-elapsed time. The program loops through a math equation six different times (50, 100, 200, 500, 700, 1000) by pressing the Start key without debugging the (F5) function. By using the "call GetMseconds" function, we can get the number of milliseconds that have elapsed since midnight.

The CPU elapsed time differs amongst the four laptops, indicating that the CPU performs differently. For the 50 loops, the Dell Inspiron 16 and ASUS Vivobook 15 show similar results, but Dell is slightly better than ASUS. The Huawei Matebook D15 has shown good performance at 200 loops, 500 loops, 700 loops, and 1000 loops. It indicates the least CPU elapsed time, the CPU is more efficient. The Lenovo IdeaPad Slim 5 consistently shows slightly higher CPU elapsed time than other laptops across all loop counts.

In conclusion, the Dell Inspiron 16 is the most efficient at 50 loops and 100 loops, but the Huawei Matebook D15 demonstrates superior performance at higher loop counts (200, 500, 700, and 1000). From the analysis above, we can say that the Huawei Matebook D15 offers the most consistent performance throughout all loop counts. At the same time the Dell Inspiron 16 and ASUS Vivobook 15 provide specific efficiency in a particular loop range. The Lenovo IdeaPad performs well, but overall it is slightly behind the other three laptops.

4.0 Conclusion and Reflection

4.1 Conclusion

After we have done the project part 1 knowing the features of the laptops together with part 2 for the coding and implementation, we are going to make a conclusion on what we get throughout this project. Most important is the analysis of the CPU elapsed times for four of our laptops, the **ASUS Vivobook, Dell Inspiron, Huawei Matebook and LENOVO 83BG**. By analyzing and comparing, it reveals notable differences in performance.

In the first part of this project, the **Dell Inspiron 16** stood out for its exceptional performance. Its CPU efficiently managed heavy workloads and multitasking. The **Dell Inspiron 16** featured the largest and most effective caches, the highest number of CPU threads, and optimized memory utilization under high load conditions to enhance data accessibility and reduce CPU strain, thereby enhancing overall performance. It also excelled in both single and multi-core processing power, making it ideal for users needing robust processing capabilities. Additionally, it demonstrated superior throughput for sequential read and write tasks, catering to applications requiring swift access to large files.

While the **Dell Inspiron 16** boasted advanced hardware specifications, during the second part of our project, the **Huawei Matebook D15** displayed the fastest CPU elapsed times for various loop iterations. Interestingly, the **Dell Inspiron 16**'s performance fall behind from the third loop onward. This suggests that hardware disparities, background processes, and system load during code execution could have influenced the results. Despite slight variations, all four laptops are competent and professional, performing tasks efficiently within a similar quality range effectively. Every laptop has its own strong point, we should choose a computer by considering also the budget and the other specific task requirements you need to have for your laptop for a better experience.

In the end, we would like to extend our heartfelt gratitude to our lecturer, Dr. Foad. Although we are with you for a short period, throughout this project, Dr. Foad has served as an outstanding mentor, offering guidance with his vast knowledge in guiding us. His provision of examples and additional insights has enriched our overall understanding. Dr. Foad's commitment to our learning journey is evident through his enlightening lectures, valuable resources, and timely feedback. He has addressed our queries, provided direction, and inspired us to explore more and understand more in benchmarking in each and every of the students. We are deeply appreciative of the opportunity to learn from Dr. Foad and his mentorship including the lectures. We used to watch your lecturers' videos since the first topics to get more quality knowledge from you. Thank you, Dr. Foad, for your exceptional guidance, education, and expertise. Your contributions have been essential to our success, and we feel privileged to have had you as our lecturer.

4.2 Reflection

1. Chau Ying Jia

This is my very first time delving into all the features and hardware specifications of my own laptop. When I purchased it, I didn't even explore much. I am grateful for this opportunity to gain a deeper understanding in acknowledging a laptop's performance and specifications. Now I have a better grasp of my laptop's capabilities and have learned how to compare different models for future purchases. This experience has provided me with fundamental knowledge of a device that is essential for a computing student. Additionally, I have acquired new skills in using benchmarking tools for further analysis, interpretation of features, and drawing conclusions. Overall, I did well and I hope this will help me in my future learning development.

2. Tan Zhi Ming

Through this project, I gained a comprehensive understanding of evaluating computer performance, which will be invaluable for future purchases or designing my own computer. I became proficient in using CPU-Z to measure performance metrics and learned about the roles of various hardware components, such as graphics cards and computer cores, in enhancing multitasking. Additionally, I explored the impact of different types of memory, such as RAM and storage options, on a computer's speed and capacity. This knowledge equips me with the ability to make informed decisions about upgrades and configurations, ensuring I can optimize any computer setup to meet specific needs and preferences.

3. Cheryl Cheong Kah Voon

In this project, I learned how to evaluate computer performance under various load conditions by running benchmarks using four different benchmarking tools: CPU-Z, GeekBench, Mini Tools, and Task Manager. After benchmarking on four separate laptops, we used Visual Studio 2019 to compile the identical code and acquire CPU runtime. By testing four laptops, I gained useful insights into how different machines manage computing activities, revealing that speed increased with the number of loops completed. I also discovered how a computer's specifications impact its performance. This experience helped me to gain a thorough understanding of the various computer performances that different computer configurations can produce, and I believe that this useful experience will play a part in my future work.

4. Lau Yee Wen

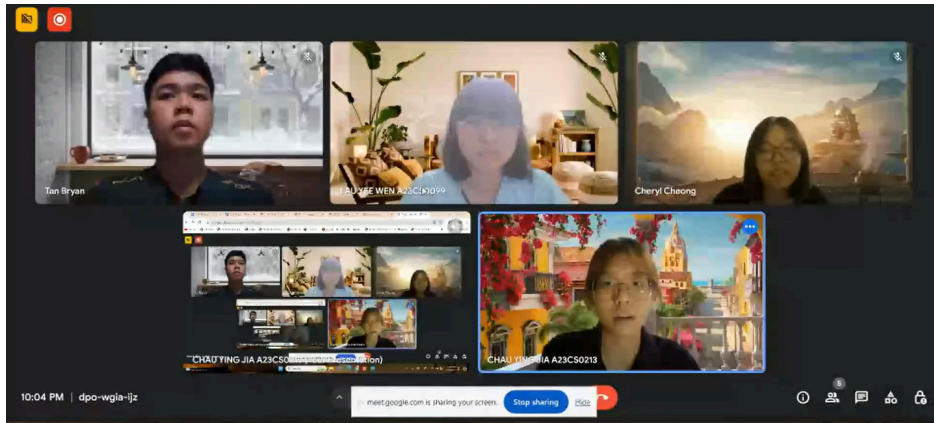
This project marked my initiation into using benchmarking tools to evaluate four distinct laptops. By employing CPU-Z, MiniTool Partition Wizard, Geekbench, and Windows Task Manager, I assessed critical CPU performance metrics such as processing speed and multi-threading efficiency. Additionally, I crafted an assembly language program to estimate and record its runtime. This hands-on experience improved my data collection and analysis skills, enhancing my ability to evaluate computer performance. The project also developed my proficiency with

benchmarking tools, data analysis, and critical thinking. These competencies are essential for optimizing hardware performance in both personal computing and professional IT environments. In summary, this project highlighted the importance of benchmarking in making informed consumer choices and optimizing IT operations. The skills and insights gained are invaluable for future personal and professional endeavors in the computing field.

References

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5. **Does the size of RAM affect speed of Computer?**
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<https://www.partitionwizard.com/download-free-from-cnet.html?p=pw&e=pw-free&r=ht>
[tps://www.partitionwizard.com/free-partition-manager.html](https://www.partitionwizard.com/free-partition-manager.html)

Appendix



1. introduction (including aim & objective) -yw
2. 2 tools features - CYJ
3. another 2 tools features- zm
4. computer specifications & part 2.0-2.2 (until flowchart) -cheryl

-comparison -

1. CPUZ - zm
2. GEEKBENCH - yj
3. WINDOWS TASK MANAGER - cheryl
4. MINITool PARTITION WIZARD - yw

-analysis-

1. CPUZ - zm
2. GEEKBENCH - yj
3. WINDOWS TASK MANAGER - cheryl
4. MINITool PARTITION WIZARD - yw

-project part 2-

1. overview & flowchart - zm
2. 2. coding& analysis- yw, cheryl
4. conclusion - yj

Slide Presentation

1. yj pg1-6, part 2 overview&flowchart, conclusion
2. zm pg7-15 2 tools comparison& analysis
3. cheryl pg16-22 another 2 tools
4. yw codes& analysis

Task Division

Video Presentation Link

: <https://youtu.be/IeZogO0Wa24>